

Joint Confidence Regions for Rankings Based on Correlated Estimates

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1 Introduction

Ranked data are commonly of interest because they allow readers to compare parameters based on ranked quantities. For example, top universities across the globe may be identified based on their institutional performance indicator; states may receive appropriate intervention according to their relative rank based on average travel times to work; and senatorial candidates who are likely to win seats can be predicted by public opinion polls prior to elections. Whenever data on the entire population are unavailable, the ranks are based on estimates rather than the unknown parameters. Consequently, their overall uncertainty—expressed through joint confidence intervals—should also be quantified. Marginally, these intervals provide information on the possible range of each rank while jointly, they facilitate comparing all ranks simultaneously rather than reporting them in isolation.

Several studies have addressed this concern through different techniques. Some approaches like those by Klein et al. (2020), Mohamad et al. (2019), Mogstad et al. (2024), Andersson et al. (1998) and Lyhagen & Ahlgren (2020), relied solely on the estimates and their standard errors, constructing joint confidence intervals either for the estimated quantities or directly for the ranks themselves. Others fit models in which institutional differences are modeled explicitly via latent effects, thereby accounting for the dependence among student observations induced by the multilevel structure (Spiegelhalter, 1996). Miller (2009) developed a bootstrap algorithm that allows for the assumption of independence in the estimators despite its potential violation while Bazylik et al. (2025) proposed a bootstrap procedure that explicitly accounts for the dependence in estimators that are multinomially distributed.

Assuming independence when constructing joint confidence regions for estimators that are, in fact, correlated may lead to overly conservative and thus

wider intervals, implying greater uncertainty—contrary to what is desired. The more general setting, which to our knowledge has not yet been explored, allows for a certain degree of correlation among the estimates to be ranked. This thesis develops a procedure capable of handling such dependencies while maintaining coverage close to the nominal level and producing relatively narrow joint confidence intervals. The proposed methodology uses only the observed estimators and their corresponding standard errors. Our proposed methodology differs from existing approaches primarily because it allows for correlation among the estimators while still controlling the nominal level. In addition, although the proposed methodology uses a parametric bootstrap—commonly used to estimate overall rank uncertainties (e.g., Andersson et al. (1998), Mogstad et al. (2024), Bazylik et al. (2025))—our implementation differs from these existing approaches. Mogstad et al. (2024) and Bazylik et al. (2025) obtained bounds for the simultaneous confidence sets which depend on quantiles from the distribution of the maximum (over k) of the differences (see Section 2.3.2, instead of just the estimators while Andersson et al. (1998) implemented it to obtain a calculated measure as discussed in Section 2.3.1.

1.1 Objective

This research builds upon Klein et al. (2020)’s methodology by extending the set of joint confidence intervals used to capture uncertainty in overall rankings. In particular, it intends to:

- Develop a procedure to construct joint confidence intervals for the ranks and the ranked parameters when the estimates to be ranked may be correlated.
- Evaluate the performance of the proposed approaches under various parameter settings.
- Compare the results of Algorithm 1 to Klein’s joint confidence regions that uses Bonferroni and independence assumption, in terms of coverage and tightness measures.

- Apply the proposed approaches to a real-life example.

1.2 Significance

In order to obtain joint confidence sets for overall ranks, Klein et al. (2020) estimates confidence intervals for the unknown parameters, with a joint coverage probability of at least $1 - \alpha$. Their goal is to produce confidence intervals that collectively produce a small difference between the upper and the lower bound to yield tighter joint uncertainty for ranks. In the same study, they used individual confidence intervals of the form $\hat{\theta}_k \pm z_{\alpha/2} \times \sigma_k$ for the population parameters $\theta_1, \theta_2, \dots, \theta_K$, assuming that the estimates of these parameters $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ are independently distributed. This approach, while simple, is no longer applicable to the case where $\hat{\theta}_k$ s may be correlated. In many cases, assuming independence in the presence of correlated quantities leads to conservative confidence intervals resulting in wider intervals which imply a higher uncertainty in overall ranks.

In the Philippines, senators are elected using the multiple non-transferable vote system (MNTV)—where candidates are voted for individually regardless of partisan membership and alliances (Ravanilla & Hicken, 2023). This applies to pre-election polls where each voter selects at most 12 from a pool of K candidates. We let the data $\mathcal{X}$ be a set of n vectors of length K , $\mathcal{X} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$, where $\mathbf{X}_i = (X_{i1}, X_{i2}, \dots, X_{iK})$ and $X_{ik} = 1$ if voter i selects candidate k and 0 otherwise. Note that $X_{ik} \sim \text{Bernoulli}(p_k)$ where p_k is the probability that voter i selects candidate k for $k \in \{1, 2, \dots, K\}$ and $i \in \{1, 2, \dots, n\}$. Public opinion polling bodies report their results by ranking the proportion of votes $\hat{\theta}_k = \frac{1}{n} \sum_{i=1}^n X_{ik}$ garnered by the K candidates from the highest to the lowest. Note that in estimating p_k , $k = 1, 2, \dots, K$, we have used its MLE $\hat{\theta}_k$. Figure 1.1 is an example of results from Pulse Asia Research, Inc. (2025); this shows that 19 out of the 64 senatorial candidates have statistical chances of winning a seat as of a week prior to the previous 2025 elections. The shaded rows correspond to the ultimate winners. Pulse Asia Inc. notes their surveys are conducted with a

95% confidence level, with confidence intervals calculated using 2.8% margin of error for a sample size of 1,200. To obtain the estimates, each response is weighted to compensate for the unequal selection probabilities in the sample design.

Candidate	Voting Preference (6-9 May 2025)	Rank
Go, Bong Go (PDPLBN)	59.5	1
Tulfo, Erwin (LAKAS)	48.7	2-4
Sotto, Tito (NPC)	44.1	2-4
Dela Rosa, Bato (PDPLBN)	42.9	2-5
Lapid, Lito (NPC)	38.0	4-8
Cayetano, Pia (NP)	34.3	5-12
Lacson, Ping (IND)	33.1	5-13
Aquino, Bam (KNP)	32.8	5-15
Bong Revilla, Ramon Jr. (LAKAS)	32.3	6-15
Binay, Abby (NPC)	30.5	6-16
Marcoleta, Rodante (IND)	28.9	6-18
Tulfo, Ben (IND)	28.9	6-19
Pacquiao, Manny Pacman (PFP)	28.4	7-19
Marcos, Imee R. (NP)	27.6	8-19
Villar, Camille (NP)	27.6	8-19
Salvador, Phillip Ipe (PDPLBN)	26.5	10-19
Pangilinan, Kiko (LP)	25.1	11-20
Revillame, Willie Wil (IND)	24.1	11-20
Abalos, Benhur (PFP)	23.7	12-20

Figure 1.1: Candidates in Contention in May 2025 Pulso ng Bayan Survey by Pulse Asia Research Inc.

In weak-party systems, candidates who belong to the same political alliance or ticket commonly co-occur in ballots and hence perform with similarity (David & Legara, 2015). As shown in Figure 1.1, the competitive list is dominated by ten out of 11 (all except Tolentino) candidates under *Alyansa para sa Bagong Pilipinas*, which was actively campaigned by the incumbent president at the time. Four candidates (Go, Dela Rosa, Salvador, and Marcoleta) are associated with the Duterte-backed alliance *DuterTen*, while two (Pangilinan and Aquino) belong to the opposition-backed *KiBam*. The remaining candidates—Marcos, Revillame, and B. Tulfo—do not formally belong to any alliance but benefit from strong name recall. This alliance-based pattern is consistent with dependence in individual voting behavior.

Because each voter is allowed to vote for up to 12 candidates, votes cast by the same individual are not independent. This dependence is reflected

at the aggregate level through the proportions of total votes received by candidates, resulting in correlated proportion estimates. Accounting for these dependencies can lead to rank estimates with reduced overall uncertainty and, consequently, shorter simultaneous confidence intervals. In particular, the range of plausible ranks for each candidate may be narrowed, depending on the degree of correlation among candidates' vote shares.

1.3 Scope and Limitations

This study proposes methodologies to construct joint confidence regions to quantify uncertainty in overall ranks, building upon the main results of Klein et al. (2020). Unlike the study of Klein et al. (2020), however, this thesis addresses the case where the parameter estimates may be correlated. Moreover, this study also proposes a methodologies to construct joint confidence regions for the ordered parameters. It applies the parametric bootstrap in implementing the procedures, with a strong emphasis on maintaining the desired coverage. However, several limitations must be acknowledged. First, the framework assumes that the data are generated from a multivariate normal distribution. Second, a limited set of correlation structures is examined to illustrate how different dependence assumptions may affect the resulting joint confidence sets. That is, we do not discuss how to estimate the correlation matrix from the data.

2 Related Literature

2.1 Ranking Problem

In the problem of estimating ranks of several unknown real-valued parameters $\theta_1, \theta_2, \dots, \theta_K$, it is desired to rank these K parameters from smallest to largest, $\theta_{(1)} < \theta_{(2)} < \dots < \theta_{(K)}$. Let $r_1, r_2, \dots, r_K$ be the true unknown ranks of $\theta_1, \theta_2, \dots, \theta_K$. For example, if $\theta_{(2)} = \theta_4$, then $r_4 = 2$. Our first goal is to derive a joint confidence region for $\theta_1, \theta_2, \dots, \theta_K$ and thereby obtain joint confidence intervals for the $r_1, r_2, \dots, r_K$ using a result from Klein et al. (2020). In deriving a joint confidence region for $\theta_1, \theta_2, \dots, \theta_K$, we will make use of the corresponding point estimates $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$. Overall uncertainty highlights that in pre-election polls, reported candidate rankings that seem decisive may in fact be much less stable once uncertainty is considered, and hence it is possible that these rankings do not align with the actual election outcomes.

Our second goal is to derive a joint confidence region for the ordered parameters $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$. In the context of senatorial elections where θ_k denotes the proportion of votes earned by a candidate, confidence intervals on ordered parameters help determine if a candidate's lead is statistically significant. If the confidence intervals for two candidates' support levels do not overlap, it is highly likely (at the specified confidence level) that the candidate with the higher point estimate is the true winner.

A mathematical definition of r_k is as follows:

$$r_k = \sum_{j=1}^K I(\theta_j \leq \theta_k) = 1 + \sum_{j:j \neq k} I(\theta_j \leq \theta_k), \quad \text{for } k = 1, 2, \dots, K. \quad (2.1)$$

Let $\hat{r}_k$ denote the estimated rank of the k th observation. This can be

computed determined based on

$$\hat{r}_k = 1 + \sum_{j:j \neq k} I(\hat{\theta}_j \leq \hat{\theta}_k), \quad \text{for } k = 1, 2, \dots, K. \quad (2.2)$$

In this chapter, we review the various approaches to solve this problem of finding a joint confidence region for $r_1, r_2, \dots, r_K$.

2.2 Klein's Joint Confidence Region for Overall Ranking Uncertainty

The study of Klein et al. (2020) assumes that $\hat{\theta}_k \sim N(\theta_k, \sigma_k^2)$, $k = 1, 2, \dots, K$ where θ_k is unknown but σ_k^2 is known. We now describe their methodology.

Suppose that for each $k \in \{1, 2, \dots, K\}$ there exists values L_k and U_k such that

$$\theta_k \in (L_k, U_k), \quad U_k, L_k \in \mathbb{R} \quad (2.3)$$

Moreover, define the quantities $I_k, \Lambda_{Lk}, \Lambda_{Rk}, \Lambda_{Ok}$ as follows:

$$\left. \begin{aligned} I_k &= \{1, 2, \dots, K\} - \{k\}, \\ \Lambda_{Lk} &= \{j \in I_k : U_j \leq L_k\}, \\ \Lambda_{Rk} &= \{j \in I_k : U_k \leq L_j\}, \\ \Lambda_{Ok} &= \{j \in I_k : U_j > L_k \text{ and } U_k > L_j\} = I_k - \{\Lambda_{Lk} \cup \Lambda_{Rk}\} \end{aligned} \right\}$$

which implies the following:

1. $j \in \Lambda_{Lk} \Leftrightarrow (L_j, U_j) \cap (L_k, U_k) = \emptyset$ and (L_j, U_j) lies to the left of (L_k, U_k) ;
2. $j \in \Lambda_{Rk} \Leftrightarrow (L_j, U_j) \cap (L_k, U_k) = \emptyset$ and (L_j, U_j) lies to the right of (L_k, U_k) ;
3. $j \in \Lambda_{Ok} \Leftrightarrow (L_j, U_j) \cap (L_k, U_k) \neq \emptyset$
4. $\Lambda_{Lk}, \Lambda_{Rk}$, and Λ_{Ok} are mutually exclusive, and $\Lambda_{Lk} \cup \Lambda_{Rk} \cup \Lambda_{Ok} = I_k$

Let $|A|$ denote the number of elements in a set A . If the condition in (2.3) holds, the main result from Klein et al. (2020) gives a range for the value of

r_k for each $k \in \{1, 2, \dots, K\}$ as follows:

$$r_k \in \{|\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, |\Lambda_{Lk}| + 3, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1\} \quad (2.4)$$

Note that the number of elements in the range given in (2.4) is $|\Lambda_{Ok}| + 1$. Since the smaller difference between U_k and L_k leads to a smaller $|\Lambda_{Ok}|$, narrower confidence intervals for $\theta_1, \theta_2, \dots, \theta_K$ are desirable.

Suppose that for random quantities L_k and U_k the event defined in (2.3) satisfies the following probability condition:

$$P \left[\bigcap_{k=1}^K \{\theta_k \in (L_k, U_k)\} \right] \geq 1 - \alpha, \quad (2.5)$$

then, by the result of Klein et al. (2020), it also follows that

$$P \left[\bigcap_{k=1}^K \{r_k \in \{|\Lambda_{Lk}| + 1, |\Lambda_{Lk}| + 2, |\Lambda_{Lk}| + 3, \dots, |\Lambda_{Lk}| + |\Lambda_{Ok}| + 1\}\} \right] \geq 1 - \alpha. \quad (2.6)$$

Due to the assumption of normality on $\hat{\theta}_k$ as well as the fact that σ_k is assumed known, Klein et al. (2020) set the confidence intervals (L_k, U_k) for θ_k to be of the form $\hat{\theta}_k \pm t \times \sigma_k$ for $k \in \{1, 2, \dots, K\}$. One may use the Bonferroni approach to choose t and call the resulting confidence region $\mathfrak{R}_{\text{bonf}}$. If such an approach is used, the choice of t that would satisfy (2.5) is $t = z_{\alpha/2K}$. Another choice of t is one that exploits the independence assumption on $\hat{\theta}_k$; denote the associated confidence region with $\mathfrak{R}_{\text{ind}}$. Such a choice is given by $z_{\gamma/2}$ where $\gamma = 1 - (1 - \alpha)^{1/K}$. To see why this is the case, note that if we define Y_k as follows:

$$Y_k = \frac{(\hat{\theta}_k - \theta_k)}{\sigma_k} \sim N(0, 1), \quad k = 1, 2, \dots, K \quad (2.7)$$

then we get

$$\begin{aligned}
\prod_{k=1}^K P(|Y_k| \leq t) &= \prod_{k=1}^K P(\hat{\theta}_k - t \cdot \sigma_k \leq \theta_k \leq \hat{\theta}_k + t \cdot \sigma_k) \\
&= \prod_{k=1}^K 1 - [1 - (1 - \alpha)^{1/K}] \\
&= 1 - \alpha.
\end{aligned} \tag{2.8}$$

2.3 Alternative Approaches for Ranking Uncertainty

While Klein's approach provides one framework for constructing joint confidence regions for ranks, several other studies have explored related problems using different formulations or assumptions. These alternative methods vary in whether they account for dependence structures, rely on model-based estimation, or use resampling techniques such as the bootstrap.

2.3.1 Calculated Measure

Other studies with similar concern include that of Andersson et al. (1998) who suggest the use of a statistic C which quantifies the number of positions ranked parameters would on average change their order due to random variation. They calculate the measure using a bootstrap approach. Working on risk ratios p_k of K units, Andersson et al. (1998) draw B bootstrap proportions $\hat{p}_k^*$ from (2.9):

$$\hat{p}_{bk}^* \sim N\left(\hat{p}_k, \frac{\hat{p}_k(1 - \hat{p}_k)}{n_k}\right), \quad k = 1, 2, \dots, K; \quad b = 1, 2, \dots, B \tag{2.9}$$

and $\hat{p}_k$ and n_k are the sample proportion and sample size, respectively, corresponding to the k th unit. For each bootstrap iteration b , they sort the set $\hat{p}_{b1}^*, \dots, \hat{p}_{bK}^*$ to get the corresponding ranks $r_{bk}^*, k = 1, 2, \dots, K$. The difference d_{bk} between the original and bootstrap rank is then calculated as $d_{bk} = |\hat{r}_k - \hat{r}_{bk}^*|$, where $\hat{r}_k$ is the rank of $\hat{p}_k$ in the set $\{\hat{p}_1, \hat{p}_2, \dots, \hat{p}_K\}$. In turn, $\bar{d}_k$ is obtained by taking the average of d_{bk} across the bootstrap

samples. Finally, the overall measure C is calculated as the average of $\bar{d}_k$ across all K units.

This can be applied to Pulse Asia senatorial pre-elections survey where we take $\hat{p}_k$ as the estimated proportion of voters who opted for candidate k and $n = 1,200$ as the common sample size. The higher C is, the less we are able to clearly distinguish between the proportion of votes received by different candidates, rendering ranks uninformative.

2.3.2 Pairwise Difference

Mohamad et al. (2019), requiring only the mean estimates and their corresponding standard errors similar to Klein et al. (2020), apply Tukey's Honest Significant Difference (HSD) to test $H_0 : \theta_j - \theta_k = 0$ for all $j \neq k \in 1, \dots, K$ at level α . Tukey's HSD is typically used to provide simultaneous confidence statements about the differences between the means while controlling the family-wise error rate (FWER). Mohamad et al. (2019) come up with a joint confidence set for ranks expressed as

$$\left(1 + \# \left\{ j : \frac{\hat{\theta}_j - \hat{\theta}_k}{\sqrt{\sigma_j^2 + \sigma_k^2}} > q_{1-\alpha} \right\}, K - \# \left\{ j : \frac{\hat{\theta}_j - \hat{\theta}_k}{\sqrt{\sigma_j^2 + \sigma_k^2}} < -q_{1-\alpha} \right\} \right), \quad (2.10)$$

for $k = 1, 2, \dots, K$

where $q_{1-\alpha}$ is the $(1 - \alpha)$ quantile of the distribution of the studentized range,

$$\max_{j,k=1,\dots,K} \frac{|\theta_j - \theta_k|}{\sqrt{\sigma_j^2 + \sigma_k^2}}.$$

In (2.10), the notation $\#\{\cdot\}$ counts the number of pairwise hypotheses that are rejected according to Tukey's HSD, which determines the lower and upper bounds of the confidence interval for the rank of $\hat{\theta}_k$. Their method has simultaneous coverage of at least $1 - \alpha$ and exactly $1 - \alpha$ when $\theta_1 = \theta_2 = \dots = \theta_K$. However, their approach tends to be overly conservative,

showing coverage levels between 0.996 and 1.0 at a 0.90 nominal level in simulations, when the θ s differ. They also demonstrated that as the true differences increase from 0 to 0.5, the coverage quickly increases from the nominal level to 1. As a remedy, they proposed a rescaling technique that brings the coverage closer to the nominal level, though it remains conservative (e.g., from 1.0 to 0.978, from 0.998 to 0.961—at 0.90 confidence level).

Mogstad et al. (2024) presents another technique that closely resembles the procedure by Klein et al. (2020) and Mohamad et al. (2019). However, they define ranks in the opposite way (i.e., larger rank value for lower estimate). They construct the rectangular confidence region in (2.11), from the pairwise differences of the estimators $\hat{\theta}_1, \dots, \hat{\theta}_K$ and an estimator of the variance of $\hat{\theta}_j - \hat{\theta}_k$. The quantities $\hat{\theta}_1, \dots, \hat{\theta}_K$ need not be independent. Let P_k be the distribution from which $\hat{\theta}_k$ is estimated; and let $\hat{P}_k$ denote the estimate of P_k . The joint confidence intervals for $r_1, \dots, r_K$ are obtained from the joint confidence intervals of the pairwise difference:

$$C(1 - \alpha, S) = \prod_{(j,k) \in S} \left[\hat{\theta}_j - \hat{\theta}_k \pm \sqrt{\sigma_j^2 + \sigma_k^2} L^{-1}(1 - \alpha, S, \hat{P}) \right], \quad (2.11)$$

$$S \subseteq \{(j, k) \in K \times K : j \neq k\}$$

where $L^{-1}(1 - \alpha, S, P)$ is the $(1 - \alpha)$ -quantile corresponding to the CDF:

$$L(x, S, P) = P \left\{ \max_{(j,k) \in S} \frac{|\hat{\theta}_j - \hat{\theta}_k - (\theta_j - \theta_k)|}{\sqrt{\sigma_j^2 + \sigma_k^2}} \leq x \right\}. \quad (2.12)$$

They added that if the estimators $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ are jointly asymptotically normally distributed, then the quantiles $L^{-1}(1 - \alpha, S, \hat{P})$, can be computed from the limiting distributions of the max-statistics shown in (2.12), through resampling methods. Their construction of confidence sets for ranks is based on hypothesis testing. Mogstad et al. (2024) show that their approach generally leads to a confidence set that is narrower than that of Klein et al. (2020).

2.3.3 Model-Based Uncertainty

In the context of ranking education and health institutions, Goldstein & Spiegelhalter (1996) treat the estimators as latent variables. Estimators are not directly measured but inferred from observed variables. They describe performance indicator as a summary statistical measurement on an institution (e.g., schools, hospitals, medical practitioners, etc.) which is meant to be related to the quality of its functioning. An example of performance indicator is “outcome” measure such as school exam results which have been used to judge institutional effectiveness. They use multilevel models to address the hierarchical nature of data structures associated with institutional performance. In its basic form,

$$\begin{aligned}y_{ik} &= \beta_0 + u_k + e_{ik} \\ \text{var}(\hat{u}_k) &= \sigma_u^2 \\ \text{var}(\hat{e}_{ik}) &= \sigma_e^2\end{aligned}$$

where y_{ik} is the exam score for student i from school k , u_k is the residual or the k th school effect, and e_{ik} is the residual effect for student i from school j . This will yield posterior estimates $\hat{u}_j$ and $\text{var}(\hat{u}_j)$ as well as $\text{rank}(\hat{u}_j)$ and $\text{var}(\text{rank}(\hat{u}_j))$ which can be used to compare the institutions. They also visualize rank uncertainty and describe that the non-overlap of confidence intervals convey a significant difference between compared institutions (Goldstein & Healy, 1995). Their example illustrates that while the multilevel model made individual estimates more accurate, it also had the effect of making the ranks even more uncertain.

2.3.4 Accounting for Data Dependencies

Some approaches explicitly account for dependencies in the data. Zhang et al. (2013) analyze U.S. age-adjusted cancer incidence and mortality rates

across states and counties by computing individual and overall simultaneous confidence intervals for age-adjusted health index using the Monte Carlo method. Because many health conditions are age-dependent, they use age-adjusted rates to minimize the confounding effect of age differences when comparing different population groups. They also extend their method to handle cases where only the adjusted rates and confidence intervals are available, aligning it more closely with the approach of Klein et al. (2020). Mohamad et al. (2019) show their technique to result in joint confidence sets with very low coverage probabilities and which are only able to reach the nominal level when differences among the means are large enough.

Hall & Miller (2009) mention that in some use cases such as institutions ranking, dependencies can be accommodated through conditioning, similar to the above approaches. However, in genomics where data on expression levels of different genes from the same individual are generally not independent, they suggest using an “independent component” version of the bootstrap on the sample, where m -out-of- n bootstrap ($m < n$) is applied as though the ranked variables were statistically independent. They show this to perform at its best when a reasonable level of correlation is present among the variables.

Bazylik et al. (2025), in their recent study, tackle the ranking of political candidates or parties using the estimated share of support each one receives in surveys. They use the approach of Mogstad et al. (2024) but account for the dependence in the estimators through their bootstrap implementation. In their setup, they let X_k be the number of votes received by the k th party from the n voters so $\mathbf{X} = (X_1, X_2, \dots, X_K)'$ is distributed according to the multinomial distribution with parameters n and $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K)'$. Their simulations show that bootstrap-based confidence sets may have coverage probability below the nominal level despite their being excessively wide. In contrast, the finite-sample confidence sets have coverage probability at least as large as expected and may even be relatively shorter.

2.4 Asymptotic Variance

Let $\hat{\theta}_{k,n}$ be an estimator of θ_k for $k = 1, 2, \dots, K$ based on a sample of size n , and let $\hat{\theta}_{(k),n}$ be the k th ordered value among $\hat{\theta}_{1,n}, \hat{\theta}_{2,n}, \dots, \hat{\theta}_{K,n}$. The estimation may be done through standard procedures such as MLE, least squares, and so forth. Moreover, suppose $\hat{\theta}_{k,n}$ s are independent for $k = 1, 2, \dots, K$. Finally, suppose, as in the application of this thesis, that $\hat{\theta}_{k,n}$ is a sample mean (or sample proportion) and θ_k is a population mean (or population proportion) and both $\hat{\theta}_{k,n}$ and θ_k are nonnegative. In this subsection, we show that

$$V[\hat{\theta}_{(k),n}] \longrightarrow k\text{th ordered value in } \{\theta_1^2 + \sigma_1^2, \theta_2^2 + \sigma_2^2, \dots, \theta_K^2 + \sigma_K^2\} - \theta_{(k)}^2 \quad (2.13)$$

as $n \rightarrow \infty$, where $\sigma_k^2 = V(\hat{\theta}_{k,n})$, which in this study is assumed to be known. The result in (2.13) gives us an expression for the asymptotic variance of $\hat{\theta}_{(k),n}$. We shall also give an expression for an estimator of this variance.

By the definition of variance, we have

$$V(\hat{\theta}_{(k),n}) = E(\hat{\theta}_{(k),n}^2) - [E(\hat{\theta}_{(k),n})]^2 \quad (2.14)$$

We make use of Theorem 2.2 of Chen (1976) on asymptotic results for ordered estimators, which we shall now state.

Theorem (Chen, 1976). Suppose that the estimator $\hat{\delta}_{k,n}$ is a strongly consistent estimator of the parameter δ_k for $k = 1, 2, \dots, K$ and based on a random sample of size n . Moreover, suppose $\hat{\delta}_{1,n}, \hat{\delta}_{2,n}, \dots, \hat{\delta}_{K,n}$ are independent. It follows that $\hat{\delta}_{(k),n}$ is a strongly consistent estimator of $\delta_{(k)}$, where $\hat{\delta}_{(k),n}$ is the k th ordered value of $\hat{\delta}_{1,n}, \hat{\delta}_{2,n}, \dots, \hat{\delta}_{K,n}$ and $\delta_{(k)}$ is the k th ordered value of the $\delta_1, \delta_2, \dots, \delta_K$.

Since $\hat{\theta}_{k,n}$, being a sample mean, is a strongly consistent estimator of θ_k , then using the result of Chen (1976), we know that $\hat{\theta}_{(k),n}$ is a strongly consistent estimator of $\theta_{(k)}$. This implies that

$$E[\hat{\theta}_{(k),n}] \longrightarrow \theta_{(k)}. \quad (2.15)$$

Moreover, since $E(\hat{\theta}_{k,n}^2) = \theta_k^2 + \sigma_k^2$ we also have

$$E[\hat{\theta}_{(k),n}^2] \longrightarrow k\text{th ordered value in } \{\theta_1^2 + \sigma_1^2, \theta_2^2 + \sigma_2^2, \dots, \theta_K^2 + \sigma_K^2\}. \quad (2.16)$$

Using the results in (2.15) and (2.16) and noting the formula in (2.14), we get the result in (2.13). It also follows that we can estimate $V[\hat{\theta}_{(k),n}]$ as follows:

$$\widehat{V[\hat{\theta}_{(k),n}]} = k\text{th ordered value among } \{\hat{\theta}_{1,n}^2 + \sigma_1^2, \dots, \hat{\theta}_{K,n}^2 + \sigma_K^2\} - \hat{\theta}_{(k),n}^2. \quad (2.17)$$

In this subsection, we have included the subscript n in our estimators to indicate that the convergence occurs as n tends to infinity. In the rest of this thesis, we shall drop this subscript.

3 Methodology

This section introduces the proposed methodologies to obtain joint confidence intervals that can be used to quantify uncertainty for the unknown overall true ranking. It addresses the scenario when the estimates of the parameters being ranked are correlated to certain degrees. Section 3.1 develops the procedure used to compute the joint confidence region for $\theta_1, \theta_2, \dots, \theta_K$. These include a non-rank and rank-based methods. Section 3.2, on the other hand, develops the procedure to compute the joint confidence region for $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$. Section 3.3 discusses some possible correlation structures to be considered for the estimates. Sections 3.4 and 3.5 discuss how to evaluate the proposed methodology and the simulation settings to be used.

3.1 Proposed Methodology to Compute the Joint Confidence Region for the Unordered Parameters

In this section, we adopt the notations defined in Section 2.1 and 2.2. Moreover, we define $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K)'$ and assume that $\hat{\boldsymbol{\theta}} \sim N(\boldsymbol{\theta}, \boldsymbol{\Sigma})$ where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K)'$ is unknown and $\boldsymbol{\Sigma}$ is a known $K \times K$ positive definite matrix. The diagonal elements of $\boldsymbol{\Sigma}$ are $\sigma_1^2, \dots, \sigma_K^2$. We note that in the literature on inferences on the ranks, it is customary to assume that the variances are known. As previously mentioned, the setup in this thesis allows the estimates to be correlated, which is in contrast with that of Klein et al. (2020), which assumes independence among the estimates. In other words, in this thesis, the covariance matrix $\boldsymbol{\Sigma}$ need not be a diagonal matrix. Similar to the study of Klein et al. (2020), knowledge of the sampling design and estimation methodology for each population is not required; we shall

first derive simultaneous confidence intervals for $\theta_1, \theta_2, \dots, \theta_K$ of the form

$$\mathfrak{R}_1 = [\hat{\theta}_1 \pm t \times \sigma_1] \times [\hat{\theta}_2 \pm t \times \sigma_2] \times \dots \times [\hat{\theta}_K \pm t \times \sigma_K]. \quad (3.1)$$

We want the joint confidence region in (3.1) to satisfy the following probability condition:

$$P\left(\hat{\theta}_k - t \cdot \sigma_k \leq \theta_k \leq \hat{\theta}_k + t \cdot \sigma_k, \forall k = 1, 2, \dots, K\right) = 1 - \alpha. \quad (3.2)$$

Equivalently, we require

$$P\left(\max_{k=1,2,\dots,K} \left| \frac{\hat{\theta}_k - \theta_k}{\sigma_k} \right| \leq t\right) = 1 - \alpha. \quad (3.3)$$

We use the parametric bootstrap to estimate the quantile t that will be used to construct the confidence intervals while controlling the coverage to be around the nominal level. In implementing the parametric bootstrap, we sample from the multivariate normal distribution with $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\Sigma}$ as mean and variance, respectively. The idea is also conceptually similar to that of Mogstad et al. (2024) who use resampling to obtain the quantile in a different context. Algorithm 1 describes the procedure to compute the joint confidence region based on unordered estimates.

Once the confidence intervals in (3.1) have been obtained, we can then use the result of Klein et al. (2020) in (2.6) to get the lower and upper bounds on the ranks $r_k, k = 1, 2, \dots, K$. That is, we also get a joint confidence region for $r_1, r_2, \dots, r_K$.

Algorithm 1 Computing the joint confidence region for the unordered parameters

Let the data be represented by $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K)'$ and suppose that $\boldsymbol{\Sigma}$ is known

1: **for** $b = 1, 2, \dots, B$ **do**

2: Generate $\hat{\boldsymbol{\theta}}_b^* \sim N_K(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma})$ and write $\hat{\boldsymbol{\theta}}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)'$

3: Compute

$$t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{bk}^* - \hat{\theta}_k}{\sigma_k} \right|$$

4: **end for**

5: Compute the $(1 - \alpha)$ -sample quantile of $t_1^*, t_2^*, \dots, t_B^*$, call this $\hat{t}$.

6: The joint confidence region for $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K)'$ is given by

$$\mathfrak{R}_1 = [\hat{\theta}_1 \pm \hat{t} \times \sigma_1] \times [\hat{\theta}_2 \pm \hat{t} \times \sigma_2] \times \dots \times [\hat{\theta}_K \pm \hat{t} \times \sigma_K].$$

3.2 Proposed Methodology to Compute a Joint Confidence Region for the Ordered Parameters

We shall also be proposing an approach to compute simultaneous confidence intervals for the ordered parameters. Suppose that for the parameters $\theta_1, \dots, \theta_K$, the corresponding ordered values are $\theta_{(1)}, \dots, \theta_{(K)}$. Similarly, let $\hat{\theta}_{(1)}, \dots, \hat{\theta}_{(K)}$ be the ordered estimates. We now derive a joint confidence region for these ordered values, and set the region to be of the following form:

$$\mathfrak{R}_2 = [\hat{\theta}_{(1)} \pm t \times \sigma_{(1)}] \times [\hat{\theta}_{(2)} \pm t \times \sigma_{(2)}] \times \dots \times [\hat{\theta}_{(K)} \pm t \times \sigma_{(K)}], \quad (3.4)$$

where $\sigma_{(k)} = \sqrt{V(\hat{\theta}_{(k)})}$. The confidence region in (3.4) is computed such that the following condition holds:

$$P\left(\hat{\theta}_{(k)} - t \cdot \sigma_{(k)} \leq \theta_{(k)} \leq \hat{\theta}_{(k)} + t \cdot \sigma_{(k)}, \forall k = 1, 2, \dots, K\right) = 1 - \alpha, \quad (3.5)$$

which is equivalent to

$$P\left(\max_{k=1,2,\dots,K} \left| \frac{\hat{\theta}_{(k)} - \theta_{(k)}}{\sigma_{(k)}} \right| \leq t\right) = 1 - \alpha, \quad (3.6)$$

and we estimate t , which in (3.6) is the $(1 - \alpha)$ -quantile of the distribution of $\max_k \left| \frac{\hat{\theta}_{(k)} - \theta_{(k)}}{\sigma_{(k)}} \right|$, via a suitable parametric bootstrap. Since $\sigma_{(k)}$ is not known, we estimate its value. We present two approaches to estimate $\sigma_{(k)}$. The first uses results from Chen (1976) and Dudewicz (1972) to obtain an expression of the asymptotic variance of $\hat{\theta}_{(k)}$ as follows:

$$V(\hat{\theta}_{(k)}) = \text{kth ordered value among } \{\theta_1^2 + \sigma_1^2, \theta_2^2 + \sigma_2^2, \dots, \theta_K^2 + \sigma_K^2\} - \theta_{(k)}^2. \quad (3.7)$$

Consequently, we get an estimate of the asymptotic variance of $\hat{\theta}_{(k)}$, given by:

$$\widehat{V(\hat{\theta}_{(k)})} = \text{kth ordered value among } \{\hat{\theta}_1^2 + \sigma_1^2, \hat{\theta}_2^2 + \sigma_2^2, \dots, \hat{\theta}_K^2 + \sigma_K^2\} - \hat{\theta}_{(k)}^2. \quad (3.8)$$

Algorithm 2 describes the procedure to compute the joint confidence region in (3.4) using this approach.

Algorithm 2 Computing the joint confidence region for the ordered parameters using the estimated asymptotic variance

1: **for** $b = 1, 2, \dots, B$ **do**

2: Generate $\hat{\boldsymbol{\theta}}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)' \sim N_K(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma})$ and let $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \dots, \hat{\theta}_{b(K)}^*$ be the corresponding ordered values

3: Compute

$$\hat{\sigma}_{b(k)}^* = \sqrt{\left[\text{kth ordered value among } \left\{ \hat{\theta}_{b1}^{*2} + \sigma_1^2, \hat{\theta}_{b2}^{*2} + \sigma_2^2, \dots, \hat{\theta}_{bK}^{*2} + \sigma_K^2 \right\} \right] - \hat{\theta}_{b(k)}^{*2}}$$

4: Compute $t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*} \right|$

5: **end for**

6: Compute the $(1 - \alpha)$ -sample quantile of $t_1^*, t_2^*, \dots, t_B^*$, call this $\hat{t}$.

7: The joint confidence region of $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ is given by

$$\mathfrak{R}_2 = \left[\hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)} \right] \times \left[\hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)} \right] \times \dots \times \left[\hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)} \right]$$

where $\hat{\sigma}_{(k)}$ is computed as

$$\hat{\sigma}_{(k)} = \sqrt{\left[\text{kth ordered value among } \left\{ \hat{\theta}_1^2 + \sigma_1^2, \hat{\theta}_2^2 + \sigma_2^2, \dots, \hat{\theta}_K^2 + \sigma_K^2 \right\} \right] - \hat{\theta}_{(k)}^2}$$

The expression for the asymptotic variance of $\hat{\theta}_{(k)}$ has actually been derived in a setting where the $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ are independent. It is not clear if the expression in (3.7) holds whenever $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K$ are correlated. Since this thesis allows for the $\hat{\theta}_k$ s to be correlated, it is possible that the expression in (3.8) for the estimate of the variance of $\hat{\theta}_{(k)}$ lacks optimality. As an alternative to using the asymptotic variance, we shall also consider using the bootstrap to estimate the variance of $\hat{\theta}_{(k)}$. To this end, a second-level bootstrap can be employed to estimate the variance. Algorithm 3 illustrates the procedure to compute the joint confidence region for the ordered parameters with the use of the bootstrap to estimate the variance of the ordered estimates. As may be expected, this approach is computationally intensive.

Algorithm 3 Computing the joint confidence region for the ordered parameters using the bootstrap estimate of the variance

- 1: **for** $b = 1, 2, \dots, B$ **do**
- 2: Generate $\hat{\boldsymbol{\theta}}_b^* = (\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*)' \sim N_K(\hat{\boldsymbol{\theta}}, \boldsymbol{\Sigma})$ and let $\hat{\theta}_{b(1)}^*, \hat{\theta}_{b(2)}^*, \dots, \hat{\theta}_{b(K)}^*$ be the corresponding ordered values of $\hat{\theta}_{b1}^*, \hat{\theta}_{b2}^*, \dots, \hat{\theta}_{bK}^*$
- 3: **for** $c = 1, 2, \dots, C$ **do**
- 4: Generate $\hat{\boldsymbol{\theta}}_{bc}^{**} = (\hat{\theta}_{bc1}^{**}, \hat{\theta}_{bc2}^{**}, \dots, \hat{\theta}_{bcK}^{**}) \sim N_K(\hat{\boldsymbol{\theta}}_b^*, \boldsymbol{\Sigma})$ and let $\hat{\theta}_{bc(1)}^{**}, \hat{\theta}_{bc(2)}^{**}, \dots, \hat{\theta}_{bc(K)}^{**}$ be the corresponding ordered values of $\hat{\theta}_{bc1}^{**}, \hat{\theta}_{bc2}^{**}, \dots, \hat{\theta}_{bcK}^{**}$
- 5: Compute $\hat{\sigma}_{b(k)}^* = \sqrt{\frac{\sum_{c=1}^C (\hat{\theta}_{bc(k)}^{**} - \bar{\hat{\theta}}_{b \cdot (k)}^{**})^2}{C - 1}}$, $\bar{\hat{\theta}}_{b \cdot (k)}^{**} = \frac{1}{C} \sum_{c=1}^C \hat{\theta}_{bc(k)}^{**}$
- 6: **end for**
- 7: Compute $t_b^* = \max_{1 \leq k \leq K} \left| \frac{\hat{\theta}_{b(k)}^* - \hat{\theta}_{(k)}}{\hat{\sigma}_{b(k)}^*} \right|$
- 8: **end for**
- 9: Compute the $(1 - \alpha)$ -sample quantile of $t_1^*, t_2^*, \dots, t_B^*$, call this $\hat{t}$.
- 10: The joint confidence region of $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ is

$$\mathfrak{R}_2 = [\hat{\theta}_{(1)} \pm \hat{t} \times \hat{\sigma}_{(1)}] \times [\hat{\theta}_{(2)} \pm \hat{t} \times \hat{\sigma}_{(2)}] \times \dots \times [\hat{\theta}_{(K)} \pm \hat{t} \times \hat{\sigma}_{(K)}]$$

where $\hat{\sigma}_{(k)}$ is computed as

$$\hat{\sigma}_{(k)} = \sqrt{\frac{\sum_{b=1}^B (\hat{\theta}_{b(k)}^* - \bar{\hat{\theta}}_{\cdot (k)}^*)^2}{B - 1}}, \quad \bar{\hat{\theta}}_{\cdot (k)}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_{b(k)}^*$$

3.3 Correlation Structures

The proposed methodologies assume that $V(\hat{\boldsymbol{\theta}}) = \boldsymbol{\Sigma}$ is known. Note that we can express $\boldsymbol{\Sigma}$ as in (3.9), where $\mathbf{R}$ is the population correlation matrix.

$$\boldsymbol{\Sigma} = \boldsymbol{\Delta}^{1/2} \mathbf{R} \boldsymbol{\Delta}^{1/2}; \quad \boldsymbol{\Delta} = \text{diag} \{ \sigma_1^2, \sigma_2^2, \dots, \sigma_K^2 \}. \quad (3.9)$$

The diagonal elements of $\mathbf{\Sigma}$, which are $\sigma_k^2 = V(\hat{\theta}_k)$ for $k = 1, 2, \dots, K$, are treated as known quantities in practice. With large sample sizes, the variance estimates are stable enough that they are treated as the actual values. However, there is limited information to use as basis for the correlations among the estimates. In this thesis, we shall be assuming certain correlation structures among the $\hat{\theta}$ s.

One structure that may be used is an equicorrelation matrix shown in (3.10). This assumes that the K variables are equally correlated, i.e., that $\rho_{jk} = \rho$ where $\rho \in [-1, 1]$ for $j \neq k \in \{1, \dots, K\}$.

$$\mathbf{R}_{\text{eq}} = (1 - \rho) \mathbf{I}_K + \rho \mathbf{1}_K \mathbf{1}_K' = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}_{K \times K} \quad (3.10)$$

In a block correlation matrix $\mathbf{R}_{\text{block}}$ with G blocks, as represented by Archakova & Hansen (2020), the correlation between any two variables is determined by the block to which the two variables belong. Each diagonal block represents an equicorrelation structure within group g , denoted by

$$\mathbf{R}_{\text{eq},g} = (1 - \rho_g) \mathbf{I}_{n_g} + \rho_g \mathbf{1}_{n_g} \mathbf{1}_{n_g}'$$

where ρ_g is the within-block correlation and n_g is the number of variables in block g such that $\sum_{g=1}^G n_g = K$. The off-diagonal blocks capture between-block correlations, represented by

$$\mathbf{C}_{g'g} = \mathbf{C}_{gg'} = \rho_{gg'} \mathbf{1}_{n_g} \mathbf{1}_{n_{g'}}' \\ \text{where } g \neq g' \in \{1, \dots, G\}$$

Thus, the full block correlation matrix can be expressed as in (3.11).

$$\mathbf{R}_{\text{block}} = \begin{bmatrix} \mathbf{R}_{\text{eq},1} & \mathbf{C}_{12} & \cdots & \mathbf{C}_{1G} \\ \mathbf{C}_{21} & \mathbf{R}_{\text{eq},2} & \cdots & \mathbf{C}_{2G} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{C}_{G1} & \mathbf{C}_{G2} & \cdots & \mathbf{R}_{\text{eq},G} \end{bmatrix}_{K \times K} \quad (3.11)$$

In the context of pre-election surveys, each block may represent correlations induced by party or ticket membership, reflecting stronger associations within parties and weaker associations between them.

3.4 Evaluation

Algorithm 4 estimates the coverage probabilities associated with the proposed methodologies. The coverage probabilities correspond to the proportion of replications in which the true parameter values are contained within the confidence intervals for all K simultaneously. Moreover, the tightness of the joint confidence region that results from Algorithm 1 is assessed using three summary measures: the arithmetic mean (T_1), geometric mean (T_2), and the metric T_3 introduced by Wright (2025), as presented in Equations 3.12–3.14.

$$T_1 = \frac{1}{K} \sum_{k=1}^K |\Lambda_{Ok}| \quad (3.12)$$

$$T_2 = \left[\prod_{k=1}^K |\Lambda_{Ok}| \right]^{1/K} \quad (3.13)$$

$$T_3 = 1 - \frac{OP}{K^2} \quad (3.14)$$

In equation 3.14, $OP = K + \sum_{k=1}^K |\Lambda_{Ok}|$ denotes the total number of occupied positions in a joint confidence region out of the total number of positions K^2 ; or the sum of the differences between the upper and lower bound of the simultaneous rank intervals added by 1, for each k . Higher values of T_1 and T_2 indicate wider confidence intervals and are therefore less desirable, whereas higher values of T_3 are preferable. T_3 can range from 0, indicating no tightness, to $\frac{K-1}{K}$, implying the confidence region only contains the estimated ranking which is likely the true ranking.

Algorithm 4 Computing the coverage probability and tightness measures

For given values of Σ and $\theta_1, \theta_2, \dots, \theta_K$ (with corresponding $\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}$ for rank-based methods)

- 1: **for** replications = 1, 2, ..., 5000 **do**
 - 2: Generate $\hat{\theta} \sim N_K(\theta, \Sigma)$
 - 3: Compute the confidence region for the unordered (ordered) parameters using Algorithm 1 ($\mathfrak{R}_1$) (Algorithms 2 and 3 ($\mathfrak{R}_2$)), Bonferroni ($\mathfrak{R}_{\text{bonf}}$), and independence assumption ($\mathfrak{R}_{\text{ind}}$).
 - 4: Check if $(\theta_1, \theta_2, \dots, \theta_K) \in \mathfrak{R}_1, \mathfrak{R}_{\text{bonf}}, \mathfrak{R}_{\text{ind}}$ ($(\theta_{(1)}, \theta_{(2)}, \dots, \theta_{(K)}) \in \mathfrak{R}_2$) for the unordered parameters (ordered parameters) and compute T_1, T_2 , and T_3 .
 - 5: **end for**
 - 6: Compute the coverage in line 4 is satisfied and the average of T_1, T_2 , and T_3 . Results of Algorithm 1 ($\mathfrak{R}_1$) is compared to ($\mathfrak{R}_{\text{bonf}}$ and $\mathfrak{R}_{\text{ind}}$).
-

3.5 Simulation Settings

For the simulation settings to be used in evaluating the performance of the proposed methodologies, we shall be varying the settings for the population mean θ , population variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_K^2$, population correlation matrix $\mathbf{R}$, and number of populations being ranked K . In carrying out the simulations, a nominal level of $1 - \alpha = 0.95$ will be used. Table 3.1 summarizes the simulation settings.

Table 3.1: True θ values

K	Variance	θ	$\mathbf{R}$
10	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks

Table 3.1: True θ values (*continued*)

K	Variance	θ	R
20	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
30	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2, 25.7, 22.2, 22.7, 22.6, 21.9, 24.2, 23.6, 22, 24.5, 25	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1, 27.2, 20.9, 21.8, 21.7, 20.4, 24.5, 23.5, 20.5, 25.1, 26	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
40	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16, 29.4, 19, 20.4, 20.3, 18.2, 25, 23.3, 18.3, 26, 27.4	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2, 25.7, 22.2, 22.7, 22.6, 21.9, 24.2, 23.6, 22, 24.5, 25, 26.5, 25.1, 25.2, 27.1, 25.3, 22.8, 20.6, 21.1, 26, 26	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1, 27.2, 20.9, 21.8, 21.7, 20.4, 24.5, 23.5, 20.5, 25.1, 26, 28.6, 26.1, 26.4, 29.7, 26.5, 22.1, 18, 18.9, 27.8, 27.8	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16, 29.4, 19, 20.4, 20.3, 18.2, 25, 23.3, 18.3, 26, 27.4, 31.9, 27.7, 28.1, 33.6, 28.2, 20.9, 14.1, 15.6, 30.5, 30.5	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks

Table 3.1: True θ values (*continued*)

K	Variance	θ	R
50	Low	22.5, 24.9, 27.5, 24.3, 24.8, 23.2, 23, 21.6, 25.1, 19, 21.3, 24.1, 26.5, 26.9, 23.2, 25.8, 21.8, 21.6, 25.6, 21.2, 25.7, 22.2, 22.7, 22.6, 21.9, 24.2, 23.6, 22, 24.5, 25, 26.5, 25.1, 25.2, 27.1, 25.3, 22.8, 20.6, 21.1, 26, 26, 22.1, 21.4, 23.2, 22.3, 22.4, 23.1, 21.4, 24.8, 27, 23.4	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	Moderate	21.5, 25.7, 30.5, 24.7, 25.5, 22.6, 22.3, 19.9, 26.1, 15.2, 19.3, 24.3, 28.7, 29.3, 22.7, 27.5, 20.1, 19.9, 27.1, 19.1, 27.2, 20.9, 21.8, 21.7, 20.4, 24.5, 23.5, 20.5, 25.1, 26, 28.6, 26.1, 26.4, 29.7, 26.5, 22.1, 18, 18.9, 27.8, 27.8, 20.8, 19.5, 22.8, 21.2, 21.3, 22.5, 19.5, 25.5, 29.6, 23	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks
	High	19.9, 27, 34.9, 25.3, 26.7, 21.9, 21.3, 17.3, 27.7, 9.4, 16.2, 24.6, 31.9, 33, 21.9, 29.9, 17.7, 17.3, 29.3, 16, 29.4, 19, 20.4, 20.3, 18.2, 25, 23.3, 18.3, 26, 27.4, 31.9, 27.7, 28.1, 33.6, 28.2, 20.9, 14.1, 15.6, 30.5, 30.5, 18.8, 16.6, 22.1, 19.4, 19.6, 21.6, 16.7, 26.7, 33.5, 22.5	$\mathbf{R}_{eq}; \rho = 0.1, 0.5, 0.9$ $\mathbf{R}_{block} - 2, 3$ blocks

All correlation coefficients are set to be positive ($\rho > 0$) since it aligns better with the intended use case. The following configurations are used for $\mathbf{R}_{block}$:

Table 3.2: Configuration for $\mathbf{R}_{block}$

2 Blocks				3 Blocks			
$\mathbf{R}_{block}$	Block No.	Size	ρ	$\mathbf{R}_{block}$	Block No.	Size	ρ
B2	1	0.5K x 0.5K	0.9	UL3	1	0.2K x 0.2K	0.9
	2	0.5K x 0.5K	0.1		2	0.3K x 0.3K	0.5
					3	0.5K x 0.5K	0.1
U2	1	0.3K x 0.3K	0.9	UH3	1	0.2K x 0.2K	0.1
	2	0.7K x 0.7K	0.1		2	0.3K x 0.3K	0.5
					3	0.5K x 0.5K	0.9

Table 3.3: σ Values.

K	σ
10	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11
20	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25
30	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25, 0.15, 0.13, 0.1, 0.1, 0.24, 0.13, 0.32, 0.19, 0.27, 0.3
40	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25, 0.15, 0.13, 0.1, 0.1, 0.24, 0.13, 0.32, 0.19, 0.27, 0.3, 0.12, 0.27, 0.09, 0.12, 0.36, 0.09, 0.15, 0.16, 0.09, 0.29
50	0.14, 0.33, 0.15, 0.23, 0.07, 0.19, 0.19, 0.37, 0.32, 0.11, 0.17, 0.27, 0.24, 0.11, 0.11, 0.13, 0.16, 0.15, 0.15, 0.25, 0.15, 0.13, 0.1, 0.1, 0.24, 0.13, 0.32, 0.19, 0.27, 0.3, 0.12, 0.27, 0.09, 0.12, 0.36, 0.09, 0.15, 0.16, 0.09, 0.29, 0.16, 0.28, 0.14, 0.07, 0.2, 0.31, 0.13, 0.14, 0.31, 0.11

4 Result

The outcomes of the simulation studies are organized into Sections 4.1 and 4.2 for the unordered and ordered parameters, respectively.

4.1 Joint Confidence Region for the Unordered Parameters

The resulting joint confidence regions for the unordered parameters obtained through the proposed Algorithm 1 (Proposed Unordered) are compared to those from Klein’s independent assumption and Bonferroni approaches, which we will refer to as KWW-Ind and KWW-Bonf. This is to show the advantage of using the proposed algorithm in the presence of correlation among ranked estimates.

Figure 4.1 plots K against the difference between the observed coverage and the nominal level of 0.95. Ideally, these values should be close to 0, indicating that the coverage is near the nominal level. Values much greater than 0 indicate conservative joint confidence regions, whereas values much less than zero indicate anti-conservative behavior.

The first row of Figure 4.1 shows results for an assumed equicorrelated structure.

Its first two facets illustrate that the difference between the coverage and 0.95 tends to be greater than zero for both KWW-Ind and KWW-Bonf with moderate to high ρ , irrespective of K . In some cases, coverage reaches approximately 0.99 when the correlation is high ($\rho = 0.9$) and the number of estimates is large (e.g., $K = 50$). In contrast, Proposed Unordered, shown in the third facet, consistently produces coverage within 0.01 of the nominal level, regardless of ρ . The advantage of Algorithm 1 is therefore evident at

moderate to high levels of correlation.

For the block diagonal correlation structures shown in the second row of Figure 4.1, the improvement from Algorithm 1 becomes distinguishable when larger block sizes have higher ρ . This is particularly evident for B2 and UH3 where 50% of the estimates are highly correlated. In U2, 30% of the estimates are highly correlated, while in UL3, only 20% are highly correlated. Under milder correlation setups such as U2 and UL3, Proposed Unordered has little to no advantage over KWW-Ind and KWW-Bonf.

The improvement from Algorithm 1 is diminished for the block diagonal correlation structure compared to that of the equicorrelation structure. This is because the correlation among estimates is weaker in the block diagonal structure, where off-diagonal blocks are zero. Moreover, since KWW-Ind and KWW-Bonf become increasingly conservative as K increases and as more estimates become moderately to highly correlated, the proposed algorithm is expected to provide closer to nominal coverage in such settings.

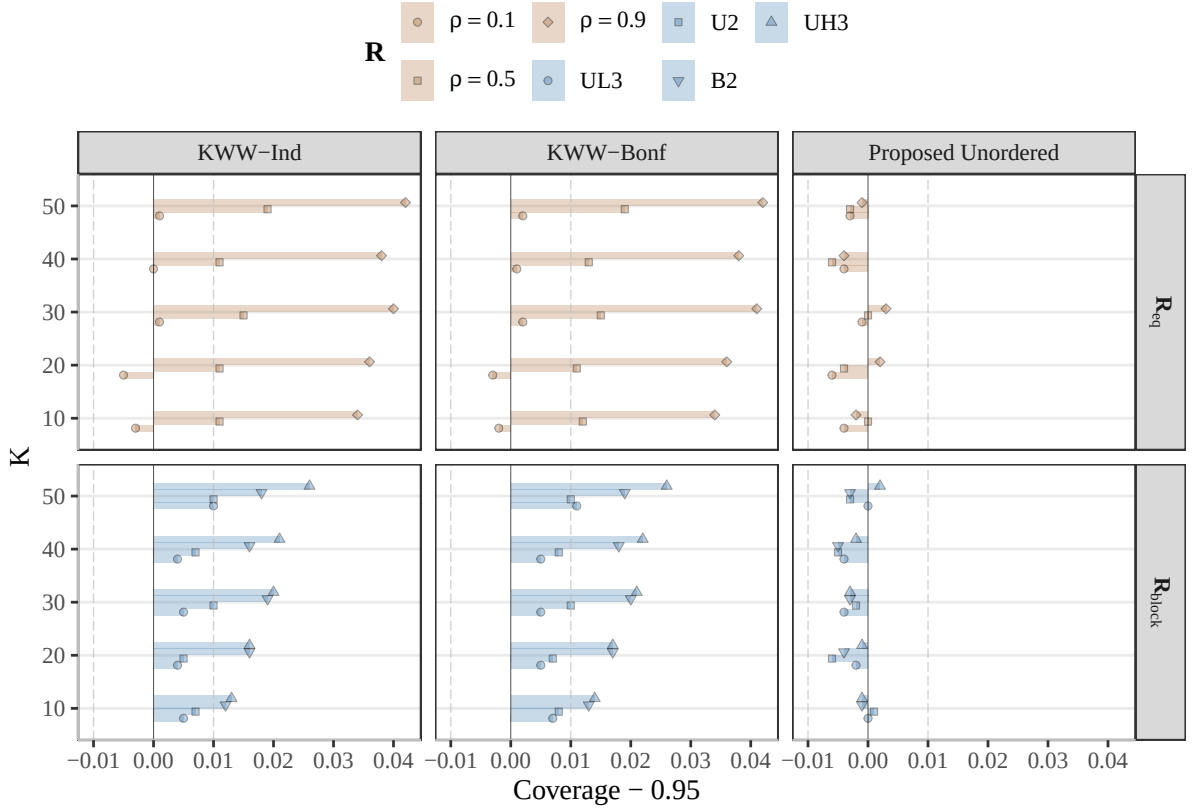


Figure 4.1: Simulation Result for Coverage of Unordered Parameters

The same coverage values are observed for different variance levels of the true parameters, $V(\theta_k)$. This results from assuming the same known Σ across cases. In Equation (3.3), it follows that the distribution of t depends only on Σ and K . This is more intuitive in the bootstrap setting as demonstrated in Algorithm 1. In step 2, we generate $\hat{\theta}_b^* \sim N_K(\hat{\theta}, \Sigma)$ and compute $\max_k \left| \frac{\hat{\theta}_{bk}^* - \hat{\theta}_k}{\sigma_k} \right|$ where $\hat{\theta}_b^* - \hat{\theta} \sim N_K(0, \Sigma)$. From Equation (3.9), this also implies that the correlation matrix $\mathbf{R}$ influences the coverage.

Results for T_1 , T_2 , and T_3 tightness measures are shown in Figure 4.2. The maximum possible value of T_3 for each K is denoted with the red dashed line. We want to observe lower values for both T_1 and T_2 and higher ones for T_3 . Altogether, these indicate that a method yields a tighter confidence region which implies less overall uncertainty.

For $\mathbf{R}_{\text{eq}}$, shown in rows 1 to 3 of Figure 4.2, T_1 , T_2 , and T_3 values indicate that Algorithm 1 produces tighter confidence region than both KWW-Ind and KWW-Bonf when ρ is moderate to high. In each facet, this is captured by the steepness of the line segments connecting the points, particularly from KWW-Bonf to Proposed Unordered. It tends to be steeper when ρ is higher. Given the same K and correlation structure, the steepness of the slope also shows that the improvement in T_1 , T_2 , and T_3 is larger with lower variance in the parameters; this is easier to observe when $\rho = 0.9$ and $K = 50$. When $\hat{\theta}$ values are more dispersed, the confidence region is narrower; however, as they vary less or become closer to each other, ranks tend to interchange more, resulting in a wider region. Hence, more room for improvement.

Moreover, while regions become less tight with higher K , as shown by the groupwise upward movement of the points from the left to the right facet for rows 1 and 2 of Figure 4.2, the increase in T_1 and T_2 is larger for parameters that vary less. This is demonstrated by the corresponding jumps of each group of same-size points across facets. As we rank more estimates, our uncertainty increases at a higher rate when the corresponding parameters are close to each other than when they are widely separated. Meanwhile, T_3 is less sensitive to K than T_1 and T_2 since it divides OP by K^2 as defined

in Equation 3.14.

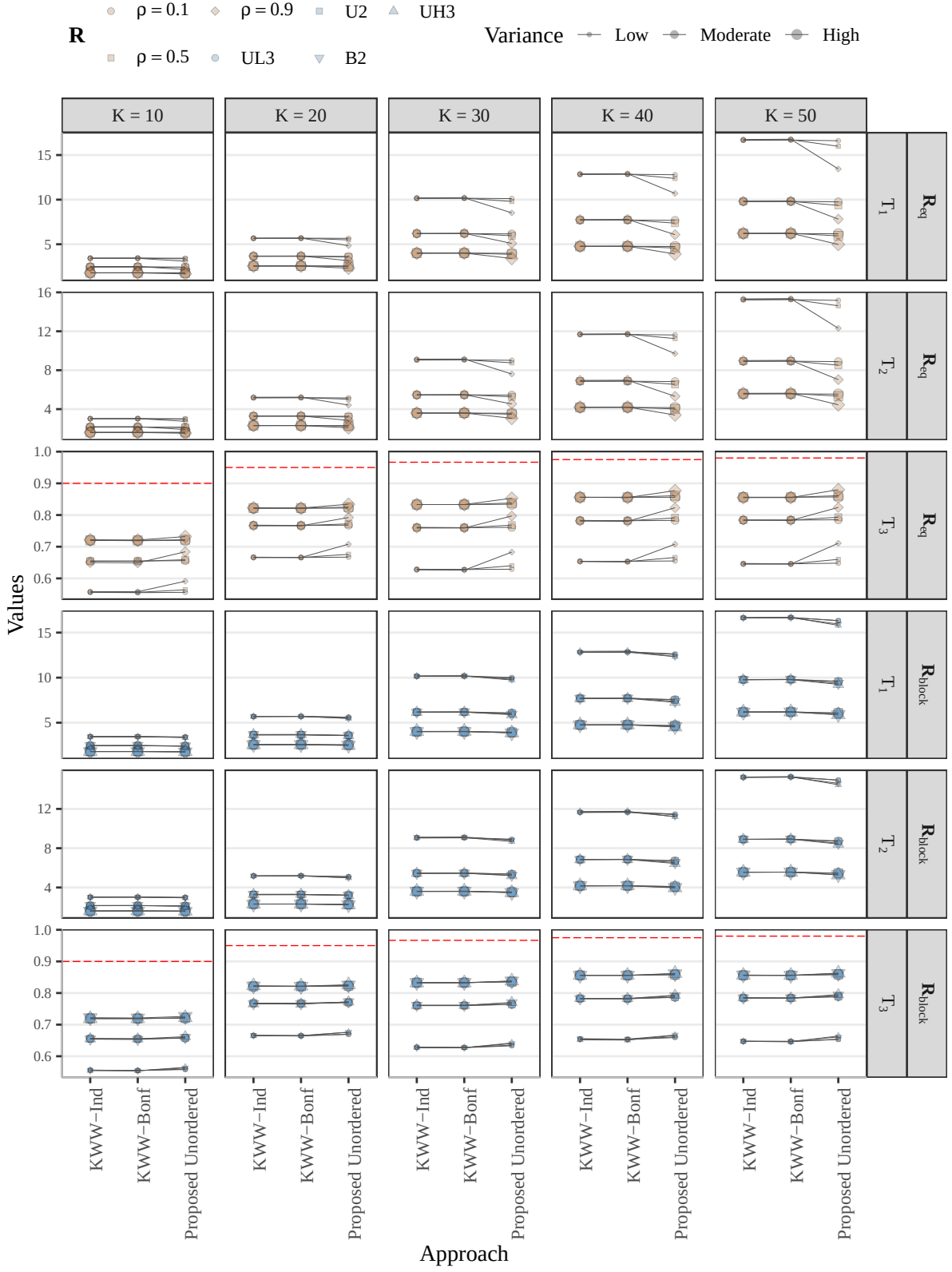


Figure 4.2: Simulation Result for Tightness Measure of Unordered Parameters

The last three rows of Figure 4.2 show the results for a block diagonal

correlation structure. Similar to the coverage results, the overall benefit of using the Proposed Unordered is reduced compared to when an equicorrelated structure is enforced. In each facet, this is illustrated through flatter slopes of line segments connecting the points corresponding to KWW-Bonf and the Proposed Unordered. Otherwise, the patterns observed for the equicorrelated structure remain consistent.

Overall, these trends show that the proposed approach is most advantageous in the presence of moderate to high correlation, especially when ranking a larger number of estimates whose underlying parameters have low variance.

The coverage values corresponding to Figure 4.1 and the values for T_1 , T_2 and T_3 in Figure 4.2 are provided in Tables A.2.1 and A.2.2–A.2.4, respectively, in the Appendix.

4.2 Joint Confidence Region for the Ordered Parameters

Figures 4.3 and 4.4 are similar to plots introduced in the previous section but show joint confidence regions for the ordered parameters namely, Proposed Asymp and Proposed Boot, constructed through Algorithms 2 and 3 respectively. With confidence regions for ranks, we are not able to map the intervals to the estimates.

The first column of Figure 4.3 illustrates how coverage for Proposed Asymp deviates from the nominal level. The deviation is almost always negative, indicating that Proposed Asymp is mostly anti-conservative. Undercoverage is largest when the parameters are widely spread, but the deviation is somewhat smaller for $K = 50$. Under high $V(\theta)$, we can also observe that lower K is associated with lower coverage for a given correlation.

Some cases with intermediate K (e.g., $K = 20$ in row 4) show smaller deviation from nominal for B2, UL3, and UH3 compared to U2. To sum up, coverage can vary with no discernible pattern across correlations and K except when $V(\theta)$ is high.

The second column of Figure 4.3 presents coverage for Proposed Boot.

It produces coverage values that do not deviate systematically from and are relatively closer to nominal when compared to Proposed Asymp. With the two proposed approaches, it is clear that Proposed Boot is superior to Proposed Asymp in terms of coverage.

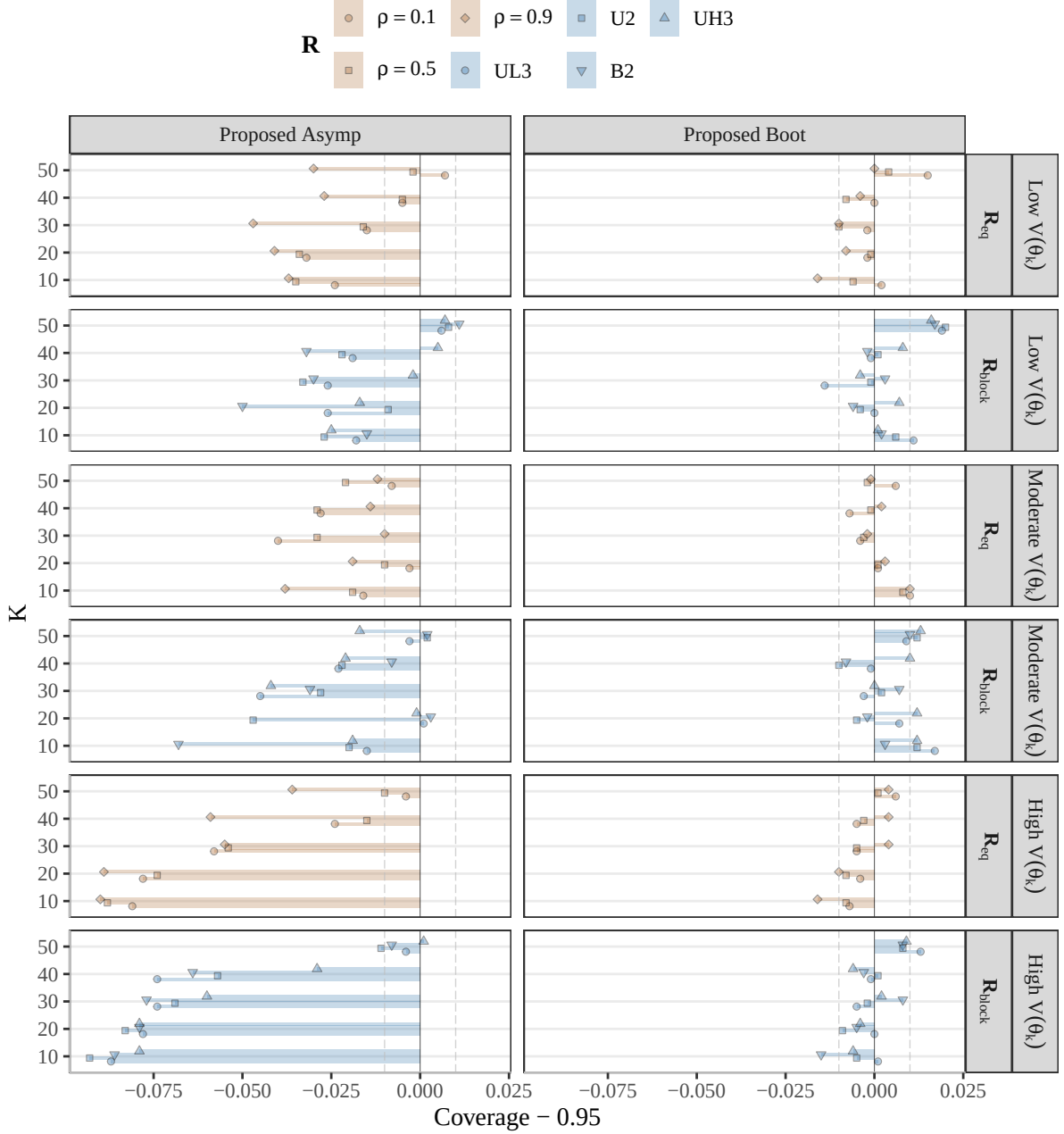


Figure 4.3: Simulation Result for Coverage of Ordered Parameters

Figure 4.4 shows the resulting T_1 for the joint confidence regions of ordered parameters. The first two columns in each facet is for Proposed Asymp while the last two is for Proposed Boot. T_2 and T_3 are not included in the plot

since they behave similarly to T_1 , except that the effect of K is less on T_3 .

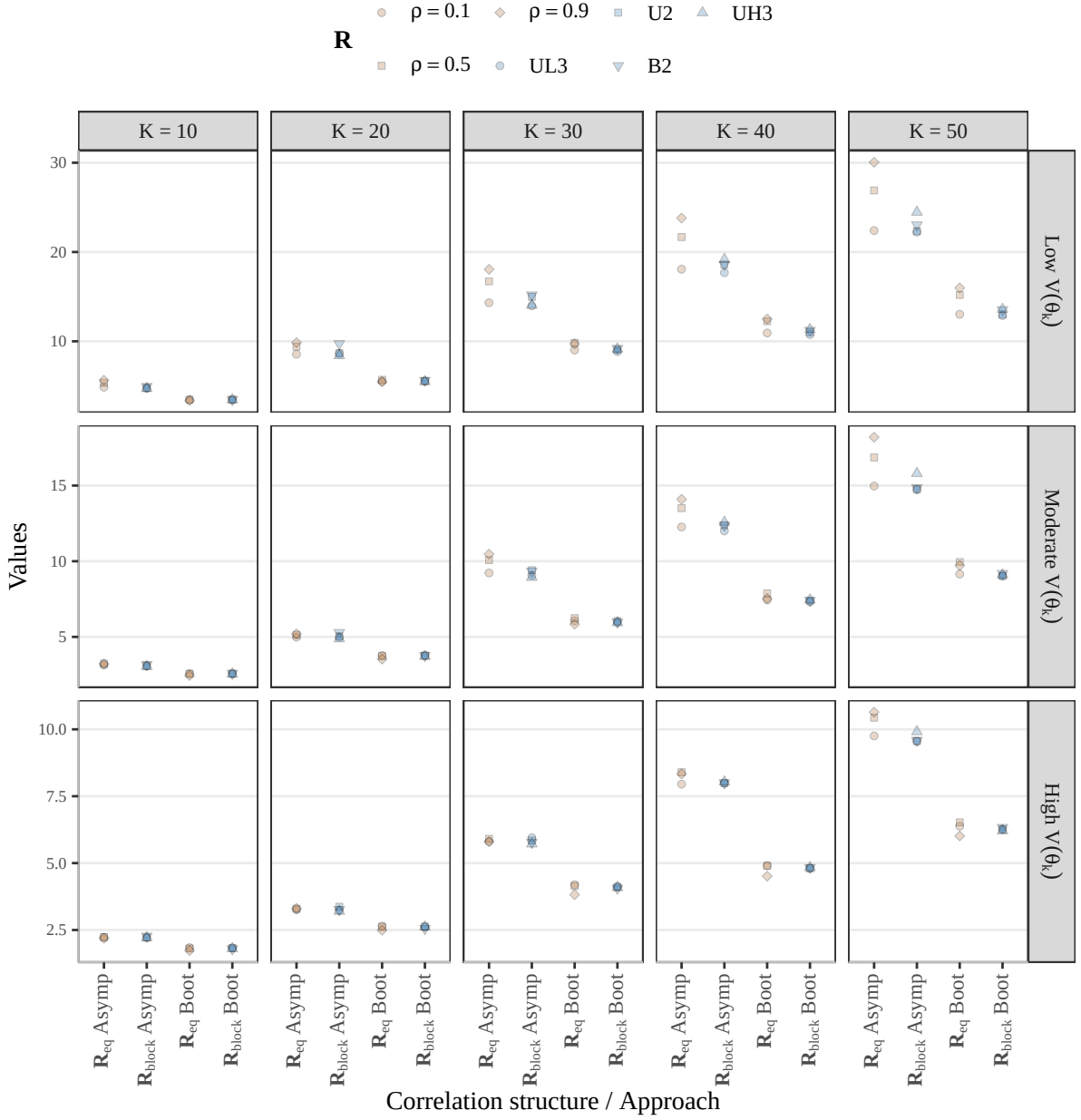


Figure 4.4: Simulation Result for Tightness Measure of Ordered Parameters

Regions become wider with larger K as shown by the upward, groupwise movement of points from left to right for T_1 in each row of Figure 4.4. T_1 values are larger with lower $V(\theta)$. This is shown through the y-axis ranges for each row that corresponds to different a level of $V(\theta)$. Across different levels of ρ in the equicorrelated structure, the difference in T_1 becomes more pronounced with larger K and lower parameter variance. For the block

correlation structure where correlation is generally weaker, these differences are less noticeable. These findings are all consistent with the outcomes for the unordered parameters.

Accounting for high correlation for Proposed Boot only helps tighten the joint confidence region when the parameters $\theta_1, \theta_2, \dots, \theta_K$ are well separated. This is easier to observe when considering the equicorrelated structure. When the parameters have a high $V(\theta)$, as shown in the third row of Figure 4.4, it is evident that the Proposed Boot joint confidence region is tightest when $\rho = 0.9$. This remains true regardless of K . However, as $V(\theta)$ increases, higher ρ tends to yield a less tight region. This behavior for higher ρ is the default for Proposed Asymp. Furthermore, Proposed Asymp still produces regions that are larger than those of Proposed Boot even with its low coverage. These results show that Proposed Boot is preferable to Proposed Asymp in terms of tightness measures.

The Appendix provides the coverage values for Figure 4.3 (Table A.3.1) and the values of T_1 , T_2 , and T_3 in Figure 4.4 (Tables A.3.2–A.3.4).

5 Application

We demonstrate the application of our proposed approaches to actual data using estimates from the Pulso ng Bayan Pre-electoral Survey (May 6–9, 2025). We also incorporate mean travel time to work, following a use case previously analyzed by Klein. In this research, we do not estimate correlation but instead assume potential sources and values. This aims to show that using an additional source of information can improve joint confidence regions by reducing overall uncertainty.

5.1 Pulse Asia Results

For the Pulse Asia results, correlations are assumed to be driven by political slates and alliances formed for the elections. To show how overall uncertainty can be reduced when accounting for these correlations, we assume a block diagonal correlation structure and let ρ vary from 0 to 0.9. This means that candidates within the same block move together, as their support levels are influenced by shared political slates and alliances. For simplicity, we assume the same ρ within all blocks. For the standard error, we use $\widehat{se} = \sqrt{\frac{\hat{p}_k(1-\hat{p}_k)}{n}}$ where $\hat{p}_k$ is the proportion of voters favoring candidate k and $n = 1,200$ is the sample size used by Pulse Asia for their survey. These are all that we need to construct the 95% joint confidence region.

The analysis considers several key blocks: Alyansa, representing candidates campaigned for by the incumbent president, and DuterTen, composed of the vice-president and former president’s allies. Imee Marcos and Camille Villar are placed in a separate block, as both are endorsed by these rival factions. The Makabayan coalition is given its own category due to its large 11-candidate lineup, while the KiBam alliance highlights the partnership between

Kiko Pangilinan and Bam Aquino. Other blocks include candidates focused on labor advocacies, independent popular senatoriables, and traditional politicians with experience in various executive departments. Finally, the remaining less prominent candidates are grouped into a single sectoral block. These make up a total of nine blocks. We construct the 95% joint confidence region and observe how T_1 , T_2 , and T_3 change.

Table 5.1: Tightness Measures for Unordered Parameters

ρ	Approach	T_1	T_2	T_3
0.0	KWW-Ind	14.882	13.115	0.689
	KWW-Bonf	14.882	13.115	0.689
		14.843	13.076	0.689
0.1		14.882	13.115	0.689
0.2		14.843	13.076	0.689
0.3		14.765	12.977	0.691
0.4	Proposed Unordered	14.765	12.977	0.691
0.5		14.686	12.894	0.692
0.6		14.608	12.788	0.694
0.7		14.686	12.894	0.692
0.8		14.294	12.434	0.700
0.9		14.294	12.434	0.700

Table 5.1 shows that higher within-block correlation leads to better T_1 , T_2 , and T_3 for the Proposed Unordered joint confidence region when compared to KWW-Ind and KWW-Bonf. This implies narrower intervals for the Proposed Unordered region, and therefore less uncertainty.

Table 5.2: Tightness Measures for Ordered Parameters

ρ	T_1		T_2		T_3	
	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
0.0	13.431	12.490	11.469	10.966	0.717	0.735
0.1	13.510	12.451	11.577	10.945	0.715	0.736
0.2	13.588	12.686	11.637	11.177	0.714	0.732
0.3	13.706	12.373	11.708	10.801	0.712	0.738
0.4	13.510	12.412	11.577	10.818	0.715	0.737
0.5	13.784	12.961	11.873	11.380	0.710	0.726
0.6	13.706	13.039	11.708	11.470	0.712	0.725
0.7	13.706	13.353	11.708	11.725	0.712	0.719
0.8	13.588	13.275	11.637	11.643	0.714	0.720
0.9	13.510	13.431	11.577	11.745	0.715	0.717

For the ordered parameters, Table 5.2 illustrates how T_1 and T_2 tend to increase while as ρ increases for the Proposed Boot joint confidence region.

Despite this behavior, it can be observed that for $\rho < 0.7$, its T_1 and T_2 values are lower than those of Proposed Asymp. Beyond that, Proposed Boot and Proposed Asymp become almost equally high. These metrics alone imply that we get more uncertain about the overall ranks when ρ is high.

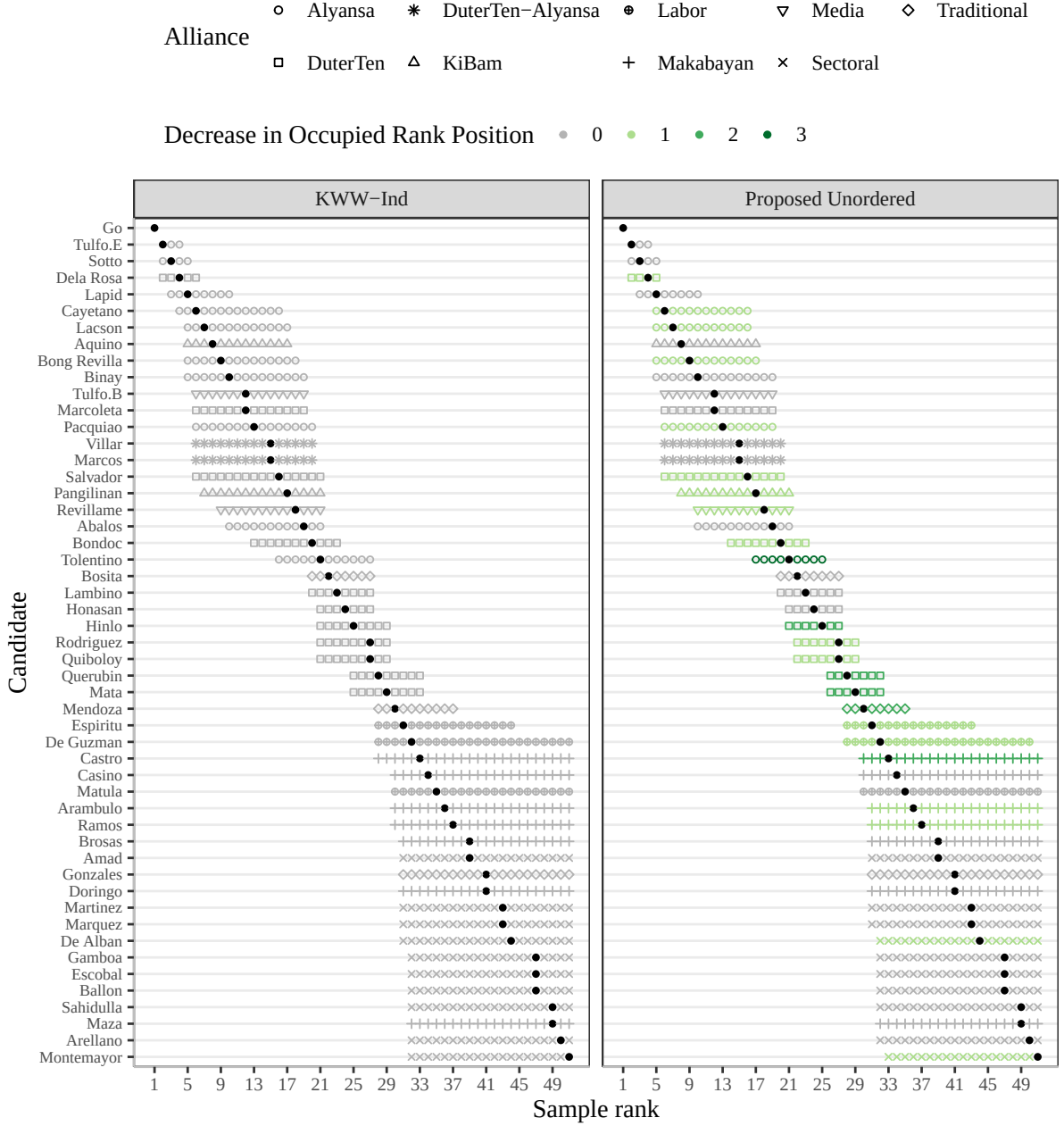


Figure 5.1: 95% Joint Confidence Region for Senatoriable Preferences

When accounting for $\rho = 0.9$, the use of Proposed Unordered region to report uncertainty leads to a 3.95% reduction in T_1 , from 14.882 to 14.294, compared to KWW-Bonf. Similar improvements are observed for T_2 and T_3 ,

with changes of 5.19% (from 13.115 to 12.434) and 1.6% (from 0.689 to 0.7), respectively.

To show how this improvement gets reflected into the joint confidence region for the unordered parameters, we use Figure 5.1 to compare the resulting KWW-Ind 95% joint confidence region in the first facet and Proposed Unordered in the second one. Note that lower rank corresponds to highly preferred candidates.

The black solid circles in the joint confidence region show the observed estimated ranking for each candidate; each specific column of the joint confidence region shows which candidates could occupy the associated rank; each specific row of the joint confidence region shows which ranks the associated candidate could occupy. For example, in the first facet, rank 3 could be occupied by Tulfo E., Sotto, Dela Rosa, and Lapid; candidate Mendoza could occupy ranks 28, 29, 30, 31, 32, $\dots$, 37. The green-shaded shapes in the second facet are the intervals that got reduced occupied positions after accounting for the correlation. The darker the shade is, the higher the reduction in occupied positions. There is no distinguishable pattern as to which among each block members get an improved interval length, but the highest improvement is seen to occur in the middle part of the estimates. KWW-Bonf is not included as it is hugely similar to KWW-Ind.

Note that for this application, we filter the candidates so that we only include those that have at least 1% of preferences garnered. This is to obtain valid results across all proposed approaches. In particular, without the filter, we cannot obtain results for Proposed Asymp. This is because of step 3 in Algorithm 2. Sampling from the normal distribution centered at $\hat{\theta}$, when $\hat{\theta}$ is close to zero, may produce negative samples of $\hat{\theta}_{bk}^*$. Since the procedure ranks the quantities $\hat{\theta}_{bk}^2 + \sigma_k^2$, negative values of $\hat{\theta}_{bk}$ can distort the ordering after squaring when coupled with a very small σ_k^2 . For instance, $\hat{\theta}_{b1}^* = -1.323539 \times 10^{-4}$ and $\hat{\theta}_{b2}^* = -0.0016393$ with corresponding $\sigma_1^2 = 2.4925 \times 10^{-6}$ and $\sigma_2^2 = 4.97 \times 10^{-6}$. Note that $\hat{\theta}_{b1}^* > \hat{\theta}_{b2}^*$ but $\hat{\theta}_{b1}^{*2} = 1.7517555 \times 10^{-8} < \hat{\theta}_{b2}^{*2} = 2.687216 \times 10^{-6}$. We compute

$\hat{\theta}_{b1}^{*2} + \sigma_1^2 = 2.5100176 \times 10^{-6}$ and $\hat{\theta}_{b2}^{*2} + \sigma_2^2 = 7.657216 \times 10^{-6}$. Taking the difference between $\left[\text{kth ordered value among } \left\{ \hat{\theta}_{b1}^{*2} + \sigma_1^2, \dots, \hat{\theta}_{bK}^{*2} + \sigma_K^2 \right\} \right]$ and $\hat{\theta}_{b(k)}^{*2}$ gives us $-1.7719841 \times 10^{-7}$ for $k = 1$. This negative difference prevents the construction of Proposed Asymp joint confidence region.

5.2 2011 Mean Travel Time to Work in the US

We employ a similar assumption of a block diagonal correlation structure induced by population density levels across states. As Klein et al. (2020) observed in their analysis of this data, states with higher population densities tend to report longer travel times to work. Five blocks are used, wherein the first has population density from 0 to 20 people per square mile, the second has 20 to 50, the third has 50 to 100, the fourth has 100 to 200, and the fifth has more than 200. We adopt a similar setup as in the Pulse Asia example.

Table 5.3: Tightness Measures for Unordered Parameters

ρ	Approach	T_1	T_2	T_3
0.0	KWW-Ind	10.765	9.166	0.769
	KWW-Bonf	10.765	9.166	0.769
		10.725	9.133	0.770
0.1		10.765	9.166	0.769
0.2		10.725	9.133	0.770
0.3		10.725	9.133	0.770
0.4	Proposed Unordered	10.647	9.068	0.772
0.5		10.569	9.004	0.773
0.6		10.294	8.813	0.779
0.7		10.216	8.722	0.780
0.8		10.098	8.616	0.782
0.9		9.627	8.036	0.792

Table 5.3 shows how T_1 , T_2 , and T_3 for Proposed Unordered start continuously improving from $\rho = 0.4$ up to 0.9, when compared to those of KWW-Bonf and KWW-Ind. This is consistent to the simulation findings in Section 4.2. We see that the largest improvement is attained with higher ρ . In the presence of $\rho = 0.9$, T_1 reduces by 10.57% to 9.627 when using Proposed Unordered from 10.765 when using KWW-Bonf to construct the joint confidence region. T_2 also improves by 12.33% from 9.166 to 8.036 while T_3 by 2.99% from 0.769 to 0.792.

Table 5.4: Tightness Measures for Ordered Parameters

ρ	T_1		T_2		T_3	
	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$
0.0	13.863	9.157	11.731	8.288	0.709	0.801
0.1	13.549	9.353	11.494	8.420	0.715	0.797
0.2	14.216	9.118	12.205	8.190	0.702	0.802
0.3	14.020	9.078	11.852	8.119	0.705	0.802
0.4	13.667	9.235	11.585	8.237	0.712	0.799
0.5	14.137	9.118	12.135	8.082	0.703	0.802
0.6	14.725	9.627	12.922	8.549	0.692	0.792
0.7	14.725	9.824	12.922	8.691	0.692	0.788
0.8	14.451	9.863	12.607	8.723	0.697	0.787
0.9	14.922	9.980	13.065	8.784	0.688	0.785

Similar to the example in the previous section, a gradual increase in T_1 and T_2 and decrease in T_3 is noticeable with higher ρ for the Proposed Boot region. However, for this dataset, Proposed Asymp joint confidence regions are hugely less tight than those of Proposed Boot.

Figure 5.2 illustrates how T_1 , T_2 , and T_3 are captured in the resulting joint confidence region for KWW-Bonf and Proposed Unordered when $\rho = 0.9$. Overall, the Proposed Unordered region is tighter. Only joint confidence regions for KWW-Bonf and Proposed Unordered are shown since the KWW-Ind result is similar to that of KWW-Bonf. Lower rank corresponds to states with lower mean travel time. It can be observed that those with no decrease in occupied rank positions originally occupy fewer rank positions (average of 7) than those whose occupied positions decreased by two to three (average of 15).

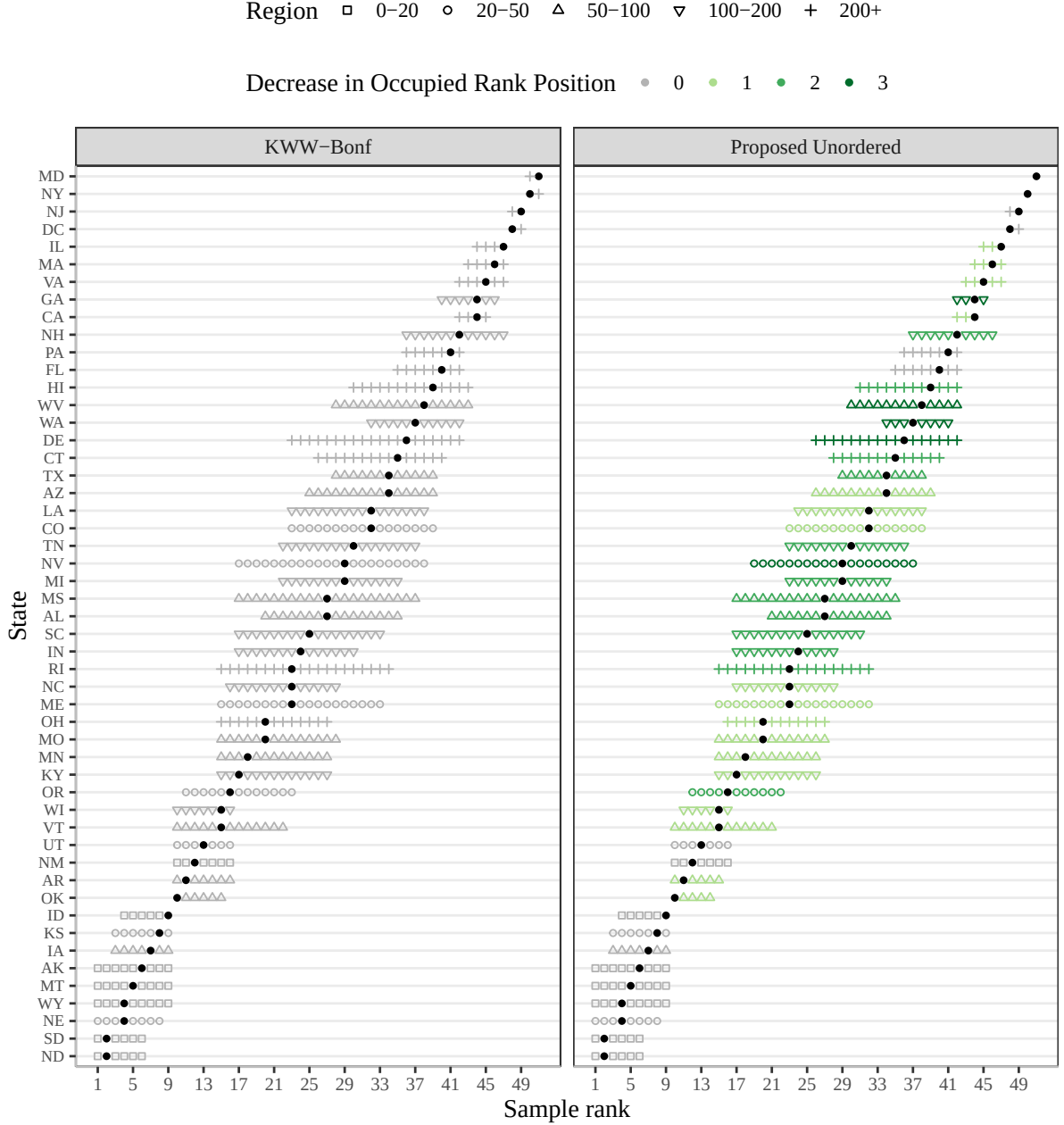


Figure 5.2: 95% Joint Confidence Region for 2011 Mean Travel Time To Work in the US

In summary, the two examples show the applicability of the proposed approaches in constructing joint confidence regions for ranks and ranked parameters. We have shown that the reduction in T_1 , T_2 , and T_3 can vary across these applications. Though a similar behavior are observed on Proposed Unordered and Proposed Boot for both the examples, Proposed Asymp is shown to yield wider regions hugely different from Proposed Boot in the second example.

6 Conclusion

In this thesis, we developed procedures for constructing joint confidence regions for both the ranks (Algorithms 2 and 3) and the ranked parameters (Algorithm 1) in settings where the estimates to be ranked may be correlated. The performance of the proposed approaches was evaluated under varying levels of parameter variance $V(\theta)$ (low, moderate, and high), different numbers of ranked parameters K , and two correlation structures with multiple correlation levels.

For the ranked parameters, we compared the joint confidence regions obtained from Algorithm 1 with KWW-Ind and KWW-Bonf. Results show that the proposed method is able to maintain coverage close to the nominal level while producing tighter joint confidence regions in the presence of moderate to high correlation. This advantage is particularly evident when ranking a larger number of estimates with low underlying parameter variance.

For the ranks, Algorithm 2 was found to systematically produce anti-conservative joint confidence regions, especially under high parameter variance. In addition, it tends to yield wider regions compared to Algorithm 3, making the latter a more reliable approach in practice.

The applicability of the proposed methods was demonstrated using the Pulse Asia and mean travel time to work datasets. The resulting joint confidence regions for the ranked parameters were consistent with the patterns observed in the simulation study, reinforcing the practical relevance of the findings.

Several limitations of the study are acknowledged. Algorithm 2 may produce a negative difference in step 3, as observed in the first example, which limits its applicability to cases when the estimates are close to zero. This issue did not arise in the simulation study, as the true parameter values

were centered at higher means. Future work may focus on addressing this limitation and improving the robustness of the method. Additionally, further calibration may be explored to achieve coverage even closer to the nominal level. Moreover, this study assumes known correlation structures and levels. An important extension would be to incorporate the estimation of correlation directly from the data. Finally, the margin of errors were held constant across parameter settings; future research may investigate how varying them could potentially affect the resulting joint confidence regions.

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Appendix

A.1 Simulation True θ s

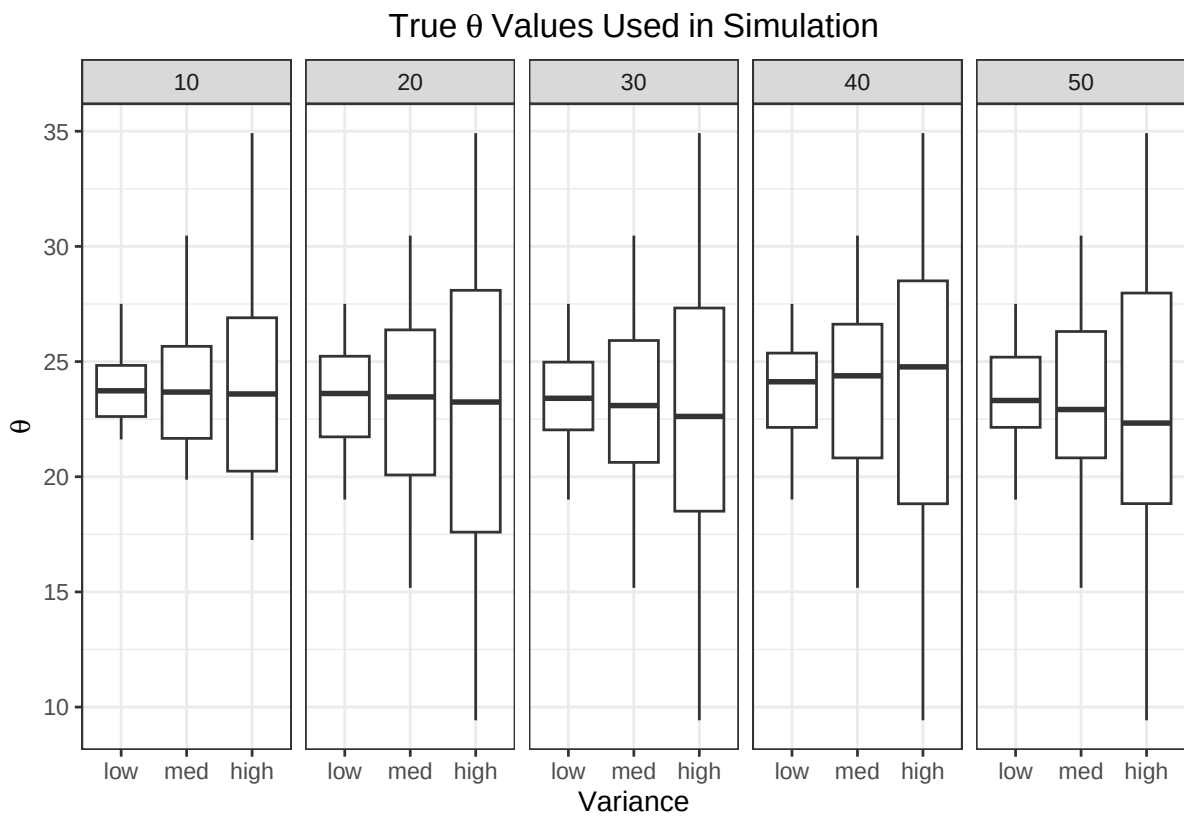


Figure A.1.1: True θ values used in the simulation have different levels of variability.

A.2 Tables: Joint Confidence Region for the Unordered Parameters

Table A.2.1: Simulation Results for Coverage of Unordered Parameters

Equicorrelated				Block Diagonal			
ρ	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$
K = 10							
0.1	0.947	0.948	0.946	UL3	0.955	0.957	0.950
0.5	0.961	0.962	0.950	U2	0.957	0.958	0.951
0.9	0.984	0.984	0.948	B2	0.962	0.963	0.949
				UH3	0.963	0.964	0.949
K = 20							
0.1	0.945	0.947	0.944	UL3	0.954	0.955	0.948
0.5	0.961	0.961	0.946	U2	0.955	0.957	0.944
0.9	0.986	0.986	0.952	B2	0.966	0.967	0.946
				UH3	0.966	0.967	0.949
K = 30							
0.1	0.951	0.952	0.949	UL3	0.955	0.955	0.946
0.5	0.965	0.965	0.950	U2	0.960	0.960	0.948
0.9	0.990	0.991	0.953	B2	0.969	0.970	0.947
				UH3	0.970	0.971	0.947
K = 40							
0.1	0.950	0.951	0.946	UL3	0.954	0.955	0.946
0.5	0.961	0.963	0.944	U2	0.957	0.958	0.945
0.9	0.988	0.988	0.946	B2	0.966	0.968	0.945
				UH3	0.971	0.972	0.948
K = 50							
0.1	0.951	0.952	0.947	UL3	0.960	0.961	0.950
0.5	0.969	0.969	0.947	U2	0.960	0.960	0.947
0.9	0.992	0.992	0.949	B2	0.968	0.969	0.947
				UH3	0.976	0.976	0.952

Table A.2.2: Simulation Results for T_1 of Unordered Parameters

		Equicorrelated									Block Diagonal										
		Low $V(\theta_k)$			Moderate $V(\theta_k)$			High $V(\theta_k)$					Low $V(\theta_k)$			Moderate $V(\theta_k)$			High $V(\theta_k)$		
K	ρ	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$		$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	$\mathfrak{R}_{Ind}$	$\mathfrak{R}_{Bonf}$	$\mathfrak{R}_1$	
10	0.1	3.444	3.451	3.436	2.446	2.451	2.440	1.803	1.805	1.802	UL3	3.441	3.448	3.417	2.445	2.449	2.429	1.802	1.804	1.795	
	0.5	3.434	3.440	3.363	2.460	2.464	2.405	1.800	1.802	1.775	U2	3.444	3.450	3.406	2.442	2.447	2.416	1.806	1.808	1.794	
	0.9	3.416	3.423	3.094	2.504	2.508	2.165	1.785	1.787	1.677	B2	3.455	3.461	3.375	2.464	2.469	2.406	1.784	1.787	1.745	
											UH3	3.454	3.460	3.351	2.446	2.451	2.379	1.809	1.811	1.780	
20	0.1	5.673	5.684	5.661	3.645	3.652	3.638	2.572	2.575	2.568	UL3	5.682	5.693	5.611	3.649	3.655	3.610	2.579	2.584	2.553	
	0.5	5.664	5.675	5.489	3.659	3.666	3.556	2.562	2.566	2.508	U2	5.700	5.711	5.605	3.637	3.642	3.584	2.573	2.578	2.539	
	0.9	5.677	5.688	4.838	3.680	3.686	3.163	2.549	2.553	2.319	B2	5.682	5.694	5.474	3.674	3.680	3.561	2.569	2.573	2.497	
											UH3	5.695	5.706	5.484	3.685	3.692	3.560	2.550	2.554	2.472	
30	0.1	10.142	10.162	10.116	6.181	6.194	6.162	4.000	4.007	3.989	UL3	10.168	10.188	10.012	6.165	6.177	6.068	4.015	4.022	3.955	
	0.5	10.148	10.167	9.797	6.194	6.207	5.959	4.002	4.009	3.868	U2	10.142	10.161	9.950	6.193	6.205	6.075	4.001	4.008	3.929	
	0.9	10.188	10.207	8.519	6.216	6.228	5.091	4.006	4.013	3.402	B2	10.179	10.197	9.804	6.171	6.183	5.933	4.011	4.019	3.868	
											UH3	10.175	10.193	9.734	6.162	6.174	5.888	4.032	4.039	3.870	
40	0.1	12.824	12.848	12.789	7.700	7.716	7.676	4.760	4.770	4.746	UL3	12.845	12.868	12.645	7.704	7.720	7.563	4.760	4.768	4.681	
	0.5	12.845	12.869	12.378	7.716	7.733	7.369	4.765	4.773	4.580	U2	12.809	12.831	12.573	7.698	7.715	7.532	4.758	4.767	4.668	
	0.9	12.882	12.902	10.697	7.767	7.784	6.076	4.775	4.784	3.902	B2	12.804	12.827	12.331	7.705	7.721	7.368	4.754	4.763	4.562	
											UH3	12.918	12.940	12.341	7.676	7.692	7.260	4.751	4.761	4.525	
50	0.1	16.649	16.682	16.602	9.780	9.800	9.754	6.188	6.200	6.172	UL3	16.634	16.666	16.337	9.783	9.803	9.602	6.185	6.196	6.075	
	0.5	16.685	16.718	15.981	9.788	9.807	9.358	6.206	6.218	5.943	U2	16.659	16.690	16.315	9.771	9.791	9.559	6.187	6.199	6.061	
	0.9	16.738	16.771	13.437	9.824	9.843	7.814	6.232	6.244	4.976	B2	16.623	16.653	15.950	9.781	9.799	9.366	6.171	6.184	5.905	
											UH3	16.654	16.687	15.806	9.765	9.782	9.254	6.186	6.198	5.872	

Table A.2.3: Simulation Results for T_2 of Unordered Parameters

K	ρ	Equicorrelated									Block Diagonal									
		Low $V(\theta_k)$			Moderate $V(\theta_k)$			High $V(\theta_k)$			Low $V(\theta_k)$			Moderate $V(\theta_k)$			High $V(\theta_k)$			
		$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	
10	0.1	3.018	3.023	3.011	2.150	2.155	2.146	1.612	1.614	1.612	UL3	3.014	3.020	2.995	2.149	2.153	2.135	1.612	1.614	1.607
	0.5	3.020	3.025	2.964	2.160	2.164	2.112	1.611	1.612	1.593	U2	3.018	3.022	2.988	2.149	2.153	2.125	1.615	1.616	1.606
	0.9	3.021	3.026	2.752	2.196	2.200	1.903	1.599	1.601	1.521	B2	3.025	3.029	2.961	2.168	2.171	2.117	1.598	1.600	1.568
											UH3	3.031	3.036	2.947	2.156	2.161	2.097	1.619	1.621	1.598
20	0.1	5.163	5.174	5.154	3.273	3.279	3.266	2.313	2.316	2.309	UL3	5.172	5.182	5.109	3.280	3.285	3.244	2.320	2.324	2.297
	0.5	5.176	5.186	5.018	3.282	3.288	3.190	2.312	2.315	2.261	U2	5.187	5.197	5.104	3.264	3.269	3.216	2.315	2.319	2.284
	0.9	5.222	5.232	4.417	3.298	3.303	2.842	2.309	2.313	2.088	B2	5.174	5.185	4.988	3.300	3.306	3.199	2.308	2.312	2.245
											UH3	5.179	5.188	4.989	3.302	3.309	3.188	2.312	2.315	2.240
30	0.1	9.041	9.059	9.018	5.458	5.470	5.442	3.588	3.595	3.579	UL3	9.058	9.075	8.920	5.450	5.460	5.367	3.605	3.611	3.550
	0.5	9.065	9.082	8.754	5.463	5.474	5.261	3.596	3.603	3.473	U2	9.038	9.055	8.868	5.468	5.479	5.365	3.590	3.597	3.525
	0.9	9.134	9.150	7.609	5.472	5.482	4.538	3.600	3.607	3.055	B2	9.125	9.141	8.802	5.445	5.454	5.244	3.612	3.619	3.476
											UH3	9.066	9.082	8.675	5.444	5.454	5.207	3.620	3.627	3.475
40	0.1	11.653	11.675	11.620	6.853	6.869	6.831	4.171	4.179	4.159	UL3	11.666	11.688	11.481	6.859	6.874	6.727	4.173	4.180	4.102
	0.5	11.681	11.704	11.250	6.873	6.889	6.545	4.183	4.190	4.017	U2	11.634	11.655	11.415	6.844	6.859	6.689	4.171	4.180	4.091
	0.9	11.717	11.735	9.707	6.956	6.972	5.318	4.210	4.217	3.395	B2	11.660	11.681	11.225	6.849	6.864	6.537	4.172	4.180	3.999
											UH3	11.735	11.755	11.200	6.837	6.852	6.446	4.169	4.178	3.968
50	0.1	15.218	15.248	15.174	8.906	8.925	8.882	5.559	5.570	5.544	UL3	15.204	15.233	14.931	8.911	8.929	8.741	5.558	5.568	5.458
	0.5	15.264	15.294	14.618	8.925	8.943	8.518	5.584	5.595	5.342	U2	15.244	15.272	14.930	8.895	8.913	8.697	5.562	5.572	5.446
	0.9	15.325	15.355	12.307	8.993	9.012	7.029	5.624	5.635	4.447	B2	15.211	15.239	14.598	8.903	8.920	8.513	5.552	5.563	5.308
											UH3	15.239	15.270	14.462	8.906	8.923	8.423	5.559	5.570	5.275

Table A.2.4: Simulation Results for T_3 of Unordered Parameters

		Equicorrelated									Block Diagonal										
		Low $V(\theta_k)$			Moderate $V(\theta_k)$			High $V(\theta_k)$					Low $V(\theta_k)$			Moderate $V(\theta_k)$			High $V(\theta_k)$		
K	ρ	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$		$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	$\Re_{Ind}$	$\Re_{Bonf}$	$\Re_1$	
10	0.1	0.556	0.555	0.556	0.655	0.655	0.656	0.720	0.719	0.720	UL3	0.556	0.555	0.558	0.656	0.655	0.657	0.720	0.720	0.720	
	0.5	0.557	0.556	0.564	0.654	0.654	0.659	0.720	0.720	0.722	U2	0.556	0.555	0.559	0.656	0.655	0.658	0.719	0.719	0.721	
	0.9	0.558	0.558	0.591	0.650	0.649	0.684	0.722	0.721	0.732	B2	0.555	0.554	0.563	0.654	0.653	0.659	0.722	0.721	0.726	
											UH3	0.555	0.554	0.565	0.655	0.655	0.662	0.719	0.719	0.722	
20	0.1	0.666	0.666	0.667	0.768	0.767	0.768	0.821	0.821	0.822	UL3	0.666	0.665	0.669	0.768	0.767	0.769	0.821	0.821	0.822	
	0.5	0.667	0.666	0.676	0.767	0.767	0.772	0.822	0.822	0.825	U2	0.665	0.664	0.670	0.768	0.768	0.771	0.821	0.821	0.823	
	0.9	0.666	0.666	0.708	0.766	0.766	0.792	0.823	0.822	0.834	B2	0.666	0.665	0.676	0.766	0.766	0.772	0.822	0.821	0.825	
											UH3	0.665	0.665	0.676	0.766	0.765	0.772	0.823	0.822	0.826	
30	0.1	0.629	0.628	0.629	0.761	0.760	0.761	0.833	0.833	0.834	UL3	0.628	0.627	0.633	0.761	0.761	0.764	0.833	0.833	0.835	
	0.5	0.628	0.628	0.640	0.760	0.760	0.768	0.833	0.833	0.838	U2	0.629	0.628	0.635	0.760	0.760	0.764	0.833	0.833	0.836	
	0.9	0.627	0.626	0.683	0.759	0.759	0.797	0.833	0.833	0.853	B2	0.627	0.627	0.640	0.761	0.761	0.769	0.833	0.833	0.838	
											UH3	0.628	0.627	0.642	0.761	0.761	0.770	0.832	0.832	0.838	
40	0.1	0.654	0.654	0.655	0.783	0.782	0.783	0.856	0.856	0.856	UL3	0.654	0.653	0.659	0.782	0.782	0.786	0.856	0.856	0.858	
	0.5	0.654	0.653	0.666	0.782	0.782	0.791	0.856	0.856	0.861	U2	0.655	0.654	0.661	0.783	0.782	0.787	0.856	0.856	0.858	
	0.9	0.653	0.652	0.708	0.781	0.780	0.823	0.856	0.855	0.877	B2	0.655	0.654	0.667	0.782	0.782	0.791	0.856	0.856	0.861	
											UH3	0.652	0.651	0.666	0.783	0.783	0.793	0.856	0.856	0.862	
50	0.1	0.647	0.646	0.648	0.784	0.784	0.785	0.856	0.856	0.857	UL3	0.647	0.647	0.653	0.784	0.784	0.788	0.856	0.856	0.858	
	0.5	0.646	0.646	0.660	0.784	0.784	0.793	0.856	0.856	0.861	U2	0.647	0.646	0.654	0.785	0.784	0.789	0.856	0.856	0.859	
	0.9	0.645	0.645	0.711	0.784	0.783	0.824	0.855	0.855	0.880	B2	0.648	0.647	0.661	0.784	0.784	0.793	0.857	0.856	0.862	
											UH3	0.647	0.646	0.664	0.785	0.784	0.795	0.856	0.856	0.863	

A.3 Tables: Joint Confidence Region for the Ordered Parameters

Table A.3.1: Simulation Results for Coverage of Ordered Parameters

		Equicorrelated						Block Diagonal							
		Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$				Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	
10	0.1	0.926	0.952	0.934	0.960	0.869	0.943	B2	0.935	0.952	0.882	0.953	0.864	0.935	
	0.5	0.915	0.944	0.931	0.958	0.862	0.942	B2	0.900	0.944	0.953	0.948	0.871	0.945	
	0.9	0.913	0.934	0.912	0.960	0.860	0.934	B2	0.920	0.953	0.919	0.957	0.873	0.958	
								B2	0.918	0.948	0.942	0.942	0.886	0.947	
20	0.1	0.918	0.948	0.947	0.951	0.872	0.946	B2	0.961	0.967	0.952	0.960	0.942	0.958	
	0.5	0.916	0.949	0.940	0.951	0.876	0.942	U2	0.923	0.956	0.930	0.962	0.857	0.945	
	0.9	0.909	0.942	0.931	0.953	0.861	0.940	U2	0.941	0.946	0.903	0.945	0.867	0.941	
								U2	0.917	0.949	0.922	0.952	0.881	0.948	
30	0.1	0.935	0.948	0.910	0.946	0.892	0.945	U2	0.928	0.951	0.928	0.940	0.893	0.951	
	0.5	0.934	0.940	0.921	0.947	0.896	0.945	U2	0.958	0.970	0.952	0.962	0.939	0.958	
	0.9	0.903	0.940	0.940	0.948	0.895	0.954	UL3	0.932	0.961	0.935	0.967	0.863	0.951	
								UL3	0.924	0.950	0.951	0.957	0.872	0.950	
40	0.1	0.945	0.950	0.922	0.943	0.926	0.945	UL3	0.924	0.936	0.905	0.947	0.876	0.945	
	0.5	0.945	0.942	0.921	0.949	0.935	0.947	UL3	0.931	0.949	0.927	0.949	0.876	0.949	
	0.9	0.923	0.946	0.936	0.952	0.891	0.954	UL3	0.956	0.969	0.947	0.959	0.946	0.963	
								UH3	0.925	0.951	0.931	0.962	0.871	0.944	
50	0.1	0.957	0.965	0.942	0.956	0.946	0.956	UH3	0.933	0.957	0.949	0.962	0.871	0.946	
	0.5	0.948	0.954	0.929	0.948	0.940	0.951	UH3	0.948	0.946	0.908	0.950	0.890	0.952	
	0.9	0.920	0.950	0.938	0.949	0.914	0.954	UH3	0.955	0.958	0.929	0.960	0.921	0.944	
								UH3	0.957	0.966	0.933	0.963	0.951	0.959	

Table A.3.2: Simulation Results for T_1 of Ordered Parameters

		Equicorrelated						Block Diagonal							
		Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$				Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	
10	0.1	4.859	3.453	3.137	2.574	2.228	1.841	UL3	4.752	3.444	3.074	2.570	2.213	1.828	
	0.5	5.352	3.460	3.236	2.559	2.235	1.809	U2	4.766	3.454	3.092	2.561	2.218	1.831	
	0.9	5.638	3.357	3.222	2.447	2.191	1.736	B2	4.935	3.483	3.168	2.591	2.268	1.793	
								UH3	4.719	3.461	3.047	2.553	2.211	1.808	
20	0.1	8.546	5.517	4.984	3.772	3.258	2.647	UL3	8.631	5.540	4.977	3.775	3.233	2.614	
	0.5	9.407	5.691	5.131	3.727	3.297	2.615	U2	8.655	5.546	5.042	3.759	3.371	2.628	
	0.9	9.852	5.450	5.201	3.533	3.307	2.493	B2	9.718	5.623	5.285	3.769	3.265	2.557	
								UH3	8.391	5.499	4.876	3.719	3.203	2.617	
30	0.1	14.319	9.019	9.226	6.058	5.811	4.193	UL3	13.977	8.851	9.058	5.965	5.950	4.127	
	0.5	16.719	9.818	10.077	6.239	5.909	4.142	U2	14.978	9.041	9.407	6.051	5.849	4.097	
	0.9	18.060	9.711	10.471	5.837	5.799	3.820	B2	15.183	9.200	9.341	5.972	5.772	4.059	
								UH3	14.046	9.138	8.935	5.942	5.716	4.093	
40	0.1	18.076	10.931	12.259	7.443	7.950	4.911	UL3	17.680	10.754	12.004	7.322	7.992	4.811	
	0.5	21.678	12.251	13.511	7.874	8.396	4.894	U2	18.580	11.019	12.383	7.404	7.980	4.800	
	0.9	23.799	12.496	14.093	7.538	8.332	4.513	B2	18.614	11.223	12.369	7.404	7.997	4.834	
								UH3	19.168	11.317	12.580	7.448	8.037	4.833	
50	0.1	22.405	13.024	14.966	9.155	9.753	6.381	UL3	22.236	12.877	14.733	9.022	9.554	6.263	
	0.5	26.895	15.196	16.858	9.943	10.435	6.522	U2	22.341	13.011	14.761	9.031	9.560	6.257	
	0.9	30.046	15.978	18.194	9.717	10.642	6.015	B2	23.029	13.533	14.860	9.184	9.587	6.331	
								UH3	24.466	13.587	15.807	9.106	9.913	6.204	

Table A.3.3: Simulation Results for T_2 of Ordered Parameters

		Equicorrelated						Block Diagonal							
		Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$				Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	
10	0.1	4.188	3.035	2.754	2.260	1.973	1.636	UL3	4.102	3.028	2.702	2.255	1.960	1.627	
	0.5	4.652	3.043	2.834	2.238	1.983	1.612	U2	4.107	3.037	2.716	2.250	1.965	1.629	
	0.9	4.968	2.969	2.810	2.126	1.953	1.559	B2	4.270	3.055	2.776	2.271	2.016	1.601	
								UH3	4.071	3.037	2.679	2.248	1.961	1.614	
20	0.1	7.738	5.083	4.530	3.391	2.927	2.390	UL3	7.812	5.102	4.523	3.398	2.904	2.368	
	0.5	8.577	5.236	4.666	3.336	2.965	2.356	U2	7.846	5.107	4.581	3.375	3.030	2.378	
	0.9	9.053	5.030	4.715	3.158	2.979	2.241	B2	8.910	5.164	4.797	3.384	2.936	2.313	
								UH3	7.596	5.055	4.425	3.343	2.885	2.374	
30	0.1	12.843	8.191	8.232	5.408	5.152	3.784	UL3	12.522	8.053	8.079	5.342	5.272	3.738	
	0.5	15.211	8.861	8.993	5.518	5.238	3.729	U2	13.485	8.209	8.387	5.395	5.185	3.705	
	0.9	16.623	8.763	9.312	5.152	5.149	3.415	B2	13.697	8.355	8.329	5.312	5.121	3.684	
								UH3	12.576	8.289	7.961	5.313	5.074	3.700	
40	0.1	16.595	10.089	11.144	6.794	7.098	4.375	UL3	16.208	9.941	10.904	6.698	7.142	4.292	
	0.5	20.211	11.226	12.299	7.124	7.520	4.326	U2	17.083	10.144	11.255	6.748	7.127	4.276	
	0.9	22.428	11.398	12.834	6.758	7.464	3.962	B2	17.110	10.365	11.255	6.751	7.146	4.306	
								UH3	17.669	10.394	11.439	6.762	7.179	4.295	
50	0.1	20.590	12.169	13.701	8.514	8.892	5.810	UL3	20.416	12.037	13.490	8.391	8.706	5.709	
	0.5	25.067	14.107	15.436	9.206	9.537	5.899	U2	20.524	12.155	13.527	8.390	8.711	5.701	
	0.9	28.350	14.781	16.655	8.951	9.729	5.399	B2	21.180	12.601	13.617	8.501	8.736	5.758	
								UH3	22.622	12.688	14.477	8.460	9.049	5.645	

Table A.3.4: Simulation Results for T_3 of Ordered Parameters

		Equicorrelated						Block Diagonal							
		Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$				Low $V(\theta_k)$		Moderate $V(\theta_k)$		High $V(\theta_k)$	
K	ρ	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$		$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	$\mathfrak{R}_{2,Asymp}$	$\mathfrak{R}_{2,Boot}$	
10	0.1	0.414	0.555	0.586	0.643	0.677	0.716	UL3	0.425	0.556	0.593	0.643	0.679	0.717	
	0.5	0.365	0.554	0.576	0.644	0.676	0.719	U2	0.423	0.555	0.591	0.644	0.678	0.717	
	0.9	0.336	0.564	0.578	0.655	0.681	0.726	B2	0.406	0.552	0.583	0.641	0.673	0.721	
								UH3	0.428	0.554	0.595	0.645	0.679	0.719	
20	0.1	0.523	0.674	0.701	0.761	0.787	0.818	UL3	0.518	0.673	0.701	0.761	0.788	0.819	
	0.5	0.480	0.665	0.693	0.764	0.785	0.819	U2	0.517	0.673	0.698	0.762	0.781	0.819	
	0.9	0.457	0.678	0.690	0.773	0.785	0.825	B2	0.464	0.669	0.686	0.762	0.787	0.822	
								UH3	0.530	0.675	0.706	0.764	0.790	0.819	
30	0.1	0.489	0.666	0.659	0.765	0.773	0.827	UL3	0.501	0.672	0.665	0.768	0.768	0.829	
	0.5	0.409	0.639	0.631	0.759	0.770	0.829	U2	0.467	0.665	0.653	0.765	0.772	0.830	
	0.9	0.365	0.643	0.618	0.772	0.773	0.839	B2	0.461	0.660	0.655	0.768	0.774	0.831	
								UH3	0.498	0.662	0.669	0.769	0.776	0.830	
40	0.1	0.523	0.702	0.669	0.789	0.776	0.852	UL3	0.533	0.706	0.675	0.792	0.775	0.855	
	0.5	0.433	0.669	0.637	0.778	0.765	0.853	U2	0.511	0.700	0.665	0.790	0.776	0.855	
	0.9	0.380	0.663	0.623	0.787	0.767	0.862	B2	0.510	0.694	0.666	0.790	0.775	0.854	
								UH3	0.496	0.692	0.661	0.789	0.774	0.854	
50	0.1	0.532	0.720	0.681	0.797	0.785	0.852	UL3	0.535	0.722	0.685	0.800	0.789	0.855	
	0.5	0.442	0.676	0.643	0.781	0.771	0.850	U2	0.533	0.720	0.685	0.799	0.789	0.855	
	0.9	0.379	0.660	0.616	0.786	0.767	0.860	B2	0.519	0.709	0.683	0.796	0.788	0.853	
								UH3	0.491	0.708	0.664	0.798	0.782	0.856	